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Characteristic analysis of occurrence probability for vortex-induced vibration of long-span bridge based on health monitoring system

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ABSTRACT

Investigating the relationship between wind climates and vortex-induced vibration (VIV) of bridges based on field measurements is fast developing currently. However, the VIV of long-span bridges clearly indicates random features. Even under the wind climates vulnerable for VIV, VIV occasionally occurs, and little attention has been paid to the non-VIV samples with the same wind condition as VIV occurs. Based on 13 years of structural health monitoring data, this study explores the key factors affecting the VIV of prototype bridge statistically. Firstly, the impact of wind speed, yaw direction, turbulence intensity and angle of attack on the VIV probability is examined synaptically. Subsequently, under the wind condition sensitive to VIV, unobserved parameters such as the damping ratio and vehicle load that are not directly observed but with strong temporal effect are discussed. The results show that the VIV can be significantly mitigated by heavy traffic volume, and the VIV probability in winter is almost negligible for high damping ratio. The fluctuation of VIV probability in one year makes short-term measurements limited in a particular season may contain large systematic errors. Finally, a prediction framework based on the Markov chain is proposed to evaluate the VIV risk under given conditions.

1. Introduction

Vortex-induced vibration (VIV) is a typical wind-induced vibration phenomenon for long-span bridges. Several bridges across the world, including the Wye bridge [1] and Kessock bridge [2] in the UK, the Trans-Tokyo Bay Crossing Bridge in Japan [3], the Deer Isle Bridge in the USA [4], the Great Belt Bridge in Denmark [5,6], the Yi Sun-Sin Bridge in Korea [7], and the Xihoumen [8,9] and the Humen Bridges [10] in China, have seen VIV incidents over time. The importance of preventing VIV in the design of bridge decks is nowadays apparent, although the self-limited load and vibration amplitude guarantee that structural safety of bridge deck is typically not harmed extremely by VIV, the oscillations could distract motorists and possibly impede traffic safety, considering possible physiological impact on vehicle drivers on the bridge. Concern should also be expressed that vibrations could convey a public stress of the bridge not being safe, resulting in a potential loss in traffic volume [5].

In addition to direct CFD calculations [11], some semi-empirical models have been utilized to qualitatively describe the nonlinear dynamic phenomena observed in VIV experiments [12–14], such as the 'lock in' region, hysteresis loops, and limit-cycle oscillation (LCO). The two most typical semi-empirical models are the wake-oscillator model [15] and the single-degree-of-freedom (SDOF) model [14]. From Scanlan's original idea, the analytical model for the 2-D sectional model should be possible to be extended to flexible prototype bridge in a simple manner by incorporating 3-D effects [14]. Besides, aerodynamic parameters obtained from the section-model study could be appropriate to be utilized to predict the VIV response of prototype bridges. However, the VIV behaviors of simplified section models in the wind tunnel tests generally show a large difference with full-scale structures. This problem is partly attributed to the existing empirical models are not able to predict the response oscillation amplitude for values of the mass and damping away from those at which their aeroelastic parameters were estimated [16], since it is quite difficult to accurately simulate the mass and damping parameters of the engineering structures in a physical wind tunnel. Above all, the complex fluid (Reynolds number effects, non-uniform inflow, turbulence, etc.) or non-fluid characteristics (Evolution of structure stiffness and damping, stochastic vehicle-induced

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vibration, etc.) in natural condition, which are generally difficult to consider in the experiment, makes the results from wind tunnel tests difficult to accurate predict VIV of prototype bridge.

With the development of bridge health monitoring system (BHMS), full-scale VIV phenomena have been investigated and discussed comprehensively based on field measurement campaigns. Smith [1] carried out the measurements of the Wye bridge in the winter of 1977–1978. The records indicated that the untoward amplitude happened when the wind velocity was between 7 m/s and 8 m/s, the wind direction was roughly perpendicular to the bridge axis, the wind inclination was between −5° and 0°, and the turbulence intensity was less than 8%. Subsequently, the wind-induced fatigue in future was evaluated on the basis of the collected short-term records. Owen et al. [2] found that the surrounding terrain could substantially influence the turbulence, which made the easterly winds at the bridge site of the Kessock bridge tend to stimulate the build up of large amplitude response than westerly winds. Larsen et al. [5] discussed the VIVs of the Great Belt East Bridge observed during the final phases of deck erection and surfacing of the suspended spans, and the guide vanes were designed and implemented to mitigate the VIV oscillations of Great Belt East Bridge. Fujino and Yoshida [3] observed that the first vertical VIV mode of a ten-span continuous single steel box girder bridge occurred in the wind direction within to the transverse axis of the bridge and the wind velocity approximately 16–17 m/s. Li et al. [9] monitored and analyzed the full-scale pressure distribution of the lower surface of bridge deck, and it was concluded that the flow separation on the lower surface of the tail, which was around the windward corner, caused large coherent vortex shedding at the leading edge. Thereafter, Li et al. [17] employed a cluster algorithm to distinguish VIVs from other vibration events, and then developed time-varying dynamic models for certain VIV events by sparse identification of nonlinear dynamics (SINDy) [18].

Considering the uncertainty of wind speed, wind direction, damping ratio, frequency and other parameters is a reasonable part of the wind-induced response analysis of wind-sensitive structures such as long-span bridges [19] and high-rise buildings [20]. Cheng et al. [19] combined probability density evolution and flutter analysis method to quantify the uncertainties in structural dynamic properties and flutter derivatives. Liu et al. [21] utilized the Gaussian Copula model to estimate the vertical, lateral and torsional buffeting responses of a bridge. However, the randomness of VIV in the field measurement of bridges has not been widely investigated. In large quantities of VIV measurement campaigns for about half a century, a core and consistent question is, how to depict the relationship of VIVs against wind parameters in a reasonable and practical manner? If the VIV of prototype bridge is understood from a deterministic perspective, it seems that a series of wind parameters can be separated from the measured wind field records to uniquely determine the amplitude and mode of VIV (e.g., [17,18]). However, because of the sparseness of the wind monitoring equipment and the impact of the previously mentioned non-fluid factors, the occurrence of VIV is often an uncertain event. As shown in Fig. 1, the established deterministic mapping from wind parameters to VIV in [17] or [18] may be inaccurate or unhelpful since it can only be used for regression analysis of VIV events that are already known to occur. Numerous samples (Even much more than the number of VIV samples) that are not excited by VIV under the same wind condition sensitive to VIV are unfortunately ignored. Cui et al. [22] attempted to model the ambient disturbance as standard Brownian motion and examine the VIV probability of Humen bridge upon Fokker–Planck–Kolmogorov equation. Limited by the number of available VIV samples form Humen Bridge (10/6/2020 to 9/10/2020), the random characteristics of VIV have not been fully revealed.

This work aims to assess the occurrence probability of VIV from the perspective of mathematical statistics. From 2009 to 2023, the wind environments and structural oscillations of the investigated bridge has been continuously accumulated by BHMS, and VIV events were observed every year. To the best of the authors' knowledge, in all long-span bridges prone to VIV, the investigated bridge may retain the largest number of complete VIV records, which provides a solid data foundation for our research. Section 2 mainly introduces the investigated bridge and corresponding wind and wind effect monitoring system. The identified VIV samples and the influence of wind parameters on VIV are discussed in Section 3, and the influence of other non-wind factors on VIV and the time-varying characteristics of VIV probability are analyzed in Section 4. Finally, in Section 5, a VIV prediction framework based on the Markov property is proposed.

Table 1
Sampling frequency variation of different sensors.

Case	Anemometers (Hz)	Accelerometers (Hz)
2009.12 to 2011.9	32	100
2011.9 to 2017.1	32	50
2017.1 to 2020.6	20	50
2020.6 to 2021.8	10	50
2021.8 to Present	5	50

2. Description of the bridge and health monitoring system

2.1. Description of the bridge

The investigated bridge, located in Zhoushan City, Zhejiang Province, China, is a two-span bridge with a 1650 m main span and a 578 m side span. As a modern long-span cable-supported bridge, the bridge connects Cezi Island in the southeast and Jintang Island in the northwest (Fig. 2). It is obvious that a hill in the lower right corner of Fig. 2 may change the local climate characteristics at the 3/4 center span position, especially when the inflow comes from the southeast. The angle between the bridge axis and the north direction is 46.5°. The bridge site is affected by the subtropical monsoon climate in the coastal area. There is more southeast monsoon in summer and more northwest wind in winter. Numerous VIVs were noticed on the bridge since the wind direction is approximately perpendicular to the longitudinal axis of bridge. The bridge deck is in the form of separated steel box girder, and the detailed geometry configuration can be found in Fig. 3.

2.2. Health monitoring system

In order to collect wind climate and wind effect information comprehensively and further provide deeper insights into wind-induced behaviors of the full-scale bridge, the health monitoring system has been implemented since 2009 and worked continuously between 2009 and 2023. Two three-dimensional ultrasonic anemometers (Young Model 8100) were installed on the southeast and northwest side of lighting columns at each quarter span location (Orange circle points in Fig. 4). It should be noted that the locations of ultrasonic anemometers installed in the 1/2 center span are near by the main cable (Fig. 5), and the inflow field are strongly disturbed by aerodynamic interference effects, so measured data at the 1/2 span would not be used in the analysis. Corresponding to anemometers, three unidirectional accelerometers were installed at the 1/4, 1/2, 3/4 central spans (Blue square points in Fig. 4). Each section had three accelerometers, two were installed symmetrically on both sides of the bridge deck to measure vertical accelerations and the horizontal distance between them was 30 m. The remained accelerometer was used to monitor the horizontal acceleration. Therefore, the lateral, vertical and torsional accelerations of the bridge deck can be thus restored by these three accelerometers. As shown in Table 1, the sampling frequencies of sensors in BHMS decreased 3 times due to data maintenance reasons, e.g., the upgrades of data storage format and sensor systems.

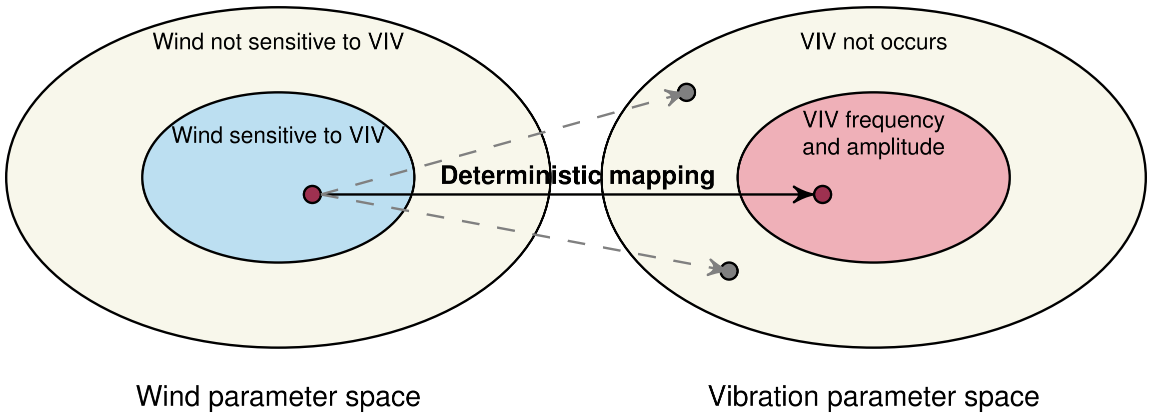


Fig. 1. Relationship between the observed wind parameters and VIV of prototype bridge.

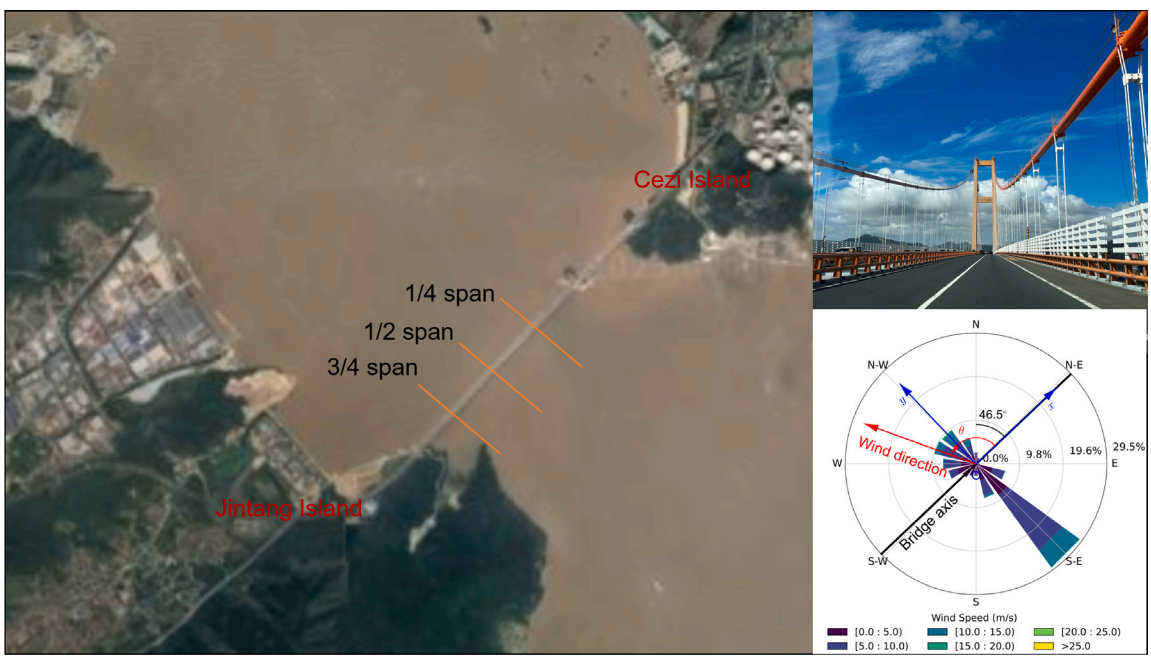


Fig. 2. Location of the investigated bridge.

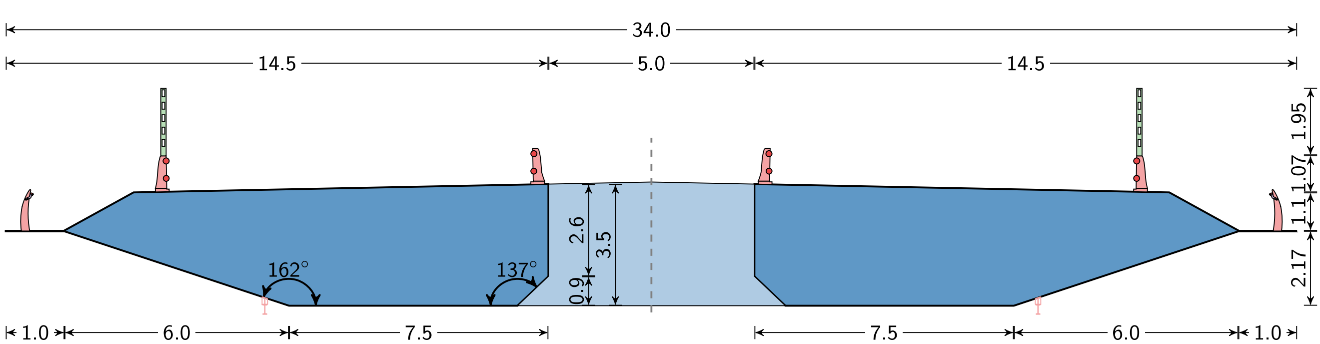


Fig. 3. Bridge Section of the investigated bridge (unit: m).

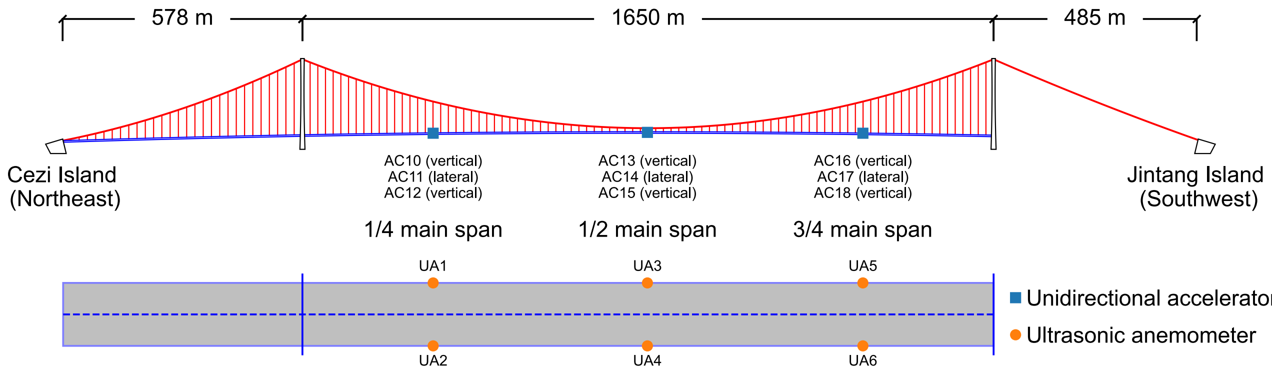


Fig. 4. Sensor location in BHMS.

Fig. 5. Anemometer at 1/2 center span.

3. Relationship between 10-min VIV samples and wind parameters

3.1. VIV identification

Different from vehicle-induced vibration and wind-induced buffeting [21], the characteristics of VIV are typical simple harmonic vibration with single frequency, and the root mean square (RMS) of acceleration response is usually larger than other vibrations [17,23]. Two features are often employed to distinguish the VIV from other vibrations in the sense of a 10 min interval:

$$\ddot{y}_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\ddot{y}_i^2} \geq 3\ \mathrm{cm/s^2} \qquad R = P_1/P_2 \geq 10 \tag{1}$$

in which $\ddot{y}_i$ is the collected acceleration sample, N is sampling number in 10-min, $\ddot{y}_{\mathrm{RMS}}$ is RMS of the measured acceleration response, and P_1 and P_2 are the maximum and the second maximum peak value of the power spectral density (PSD) of acceleration, respectively. The acceleration threshold 3 $\mathrm{cm/s^2}$ and frequency ratio 10 are approximately estimated from the previous observations. The recognition results of all datasets are shown in Table 2. In total, 3471 10-min VIV samples (about 741 VIV events, the definition of VIV events is shown later) are recognized in all the 562707 monitored datasets from 2010 to 2022, and the VIV of 0.183 Hz, 0.276 Hz and 0.327 Hz are most likely to be observed, so the following discussion will focus on these three modes.

Table 2
Frequencies and maximum RMSs of all 10-min VIV samples.

VIV frequency (Hz)	VIV number	Maximum RMS ($\mathrm{cm/s^2}$)
0.183	566	13.0
0.230	43	18.6
0.276	273	23.5
0.327	2200	41.2
0.378	98	11.8
0.436	174	40.8
0.495	117	40.9

The probability density functions $f(A = x|A_{\mathrm{VIV}}^{(f_i)})$ (PDFs) of the displacements of different modes are then fitted by truncated exponential function because of the self-limited amplitude of VIV.

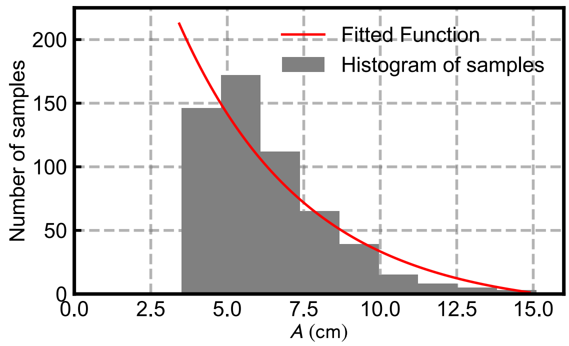
$$f(A = x|A_{\mathrm{VIV}}^{(f_i)}) = a \times \exp(-bx) + c; \quad x \in [x_1, x_2] \qquad \int_{x_1}^{x_2} f(A = x|A_{\mathrm{VIV}}^{(f_i)})\mathrm{d}x = 1 \tag{2}$$

where A here is defined as the maximum displacement of the main span, $A_{\mathrm{VIV}}^{(f_i)}$ represents the occurrence of VIV with frequency f_i and x_1 and x_2 means the minimum and maximum observed amplitudes, respectively. The fitted PDFs are shown in Fig. 6. The tails of the PDFs of 0.276 Hz and 0.327 Hz VIV are relatively flat, which means that these two modes reach the limit cycle state more frequently compared with the 0.183 Hz VIV.

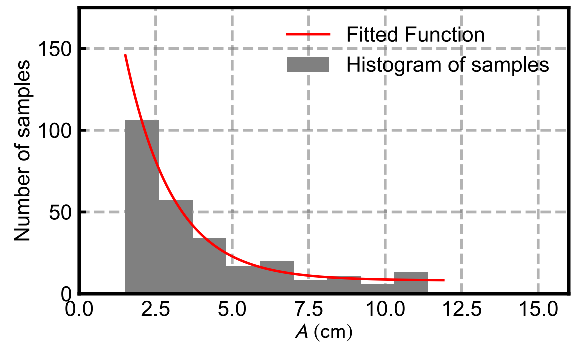
Fig. 7 plots two whole VIV events, which refer to the ordered time series composed of identified VIV samples that are sufficient close (less than an hour) in time domain. The red line in Fig. 7 represents the acceleration of VIV samples, and the gray line is the acceleration of non-VIV samples. For the VIV event in Fig. 7(a), the bridge deck was always in a simple harmonic motion from the beginning to the end of the VIV, and there was a clear process of amplitude from growth to recession. Different from that, the VIV event in Fig. 7(b) had an irregular amplitude. Since the VIV samples often appeared alternately with the non-VIV samples, the continuous VIV time was short, showing the behavior of intermittent vibration. In addition, the amplitudes of the LCO of these two VIV events show strong fluctuations, and the stationary amplitudes are difficult to be observed and determined.

3.2. Basic wind climates of VIV

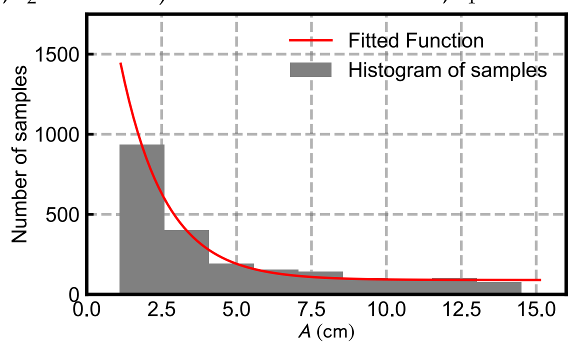
Fig. 8 plots the maximum RMS and mode of the vibration samples against the wind speed of all available records. U is the 10 min average wind speed, θ is the yaw angle between the average wind speed and the bridge axis (the local coordinate is defined in Fig. 2).



(a) 0.183 Hz ($a = 0.775$ cm^{-1}, $b = 0.24$ cm^{-1}, $c = -0.020$ cm^{-1}, $x_1 = 3.7$ cm, $x_2 = 13.6$ cm)



(b) 0.276 Hz ($a = 2.032$ cm^{-1}, $b = 0.64$ cm^{-1}, $c = -0.046$ cm^{-1}, $x_1 = 1.5$ cm, $x_2 = 11.9$ cm)



(c) 0.327 Hz ($a = 0.914$ cm^{-1}, $b = 0.67$ cm^{-1}, $c = 0.029$ cm^{-1}, $x_1 = 1.2$ cm, $x_2 = 14.8$ cm)

Fig. 6. Distributions of displacement amplitude A.

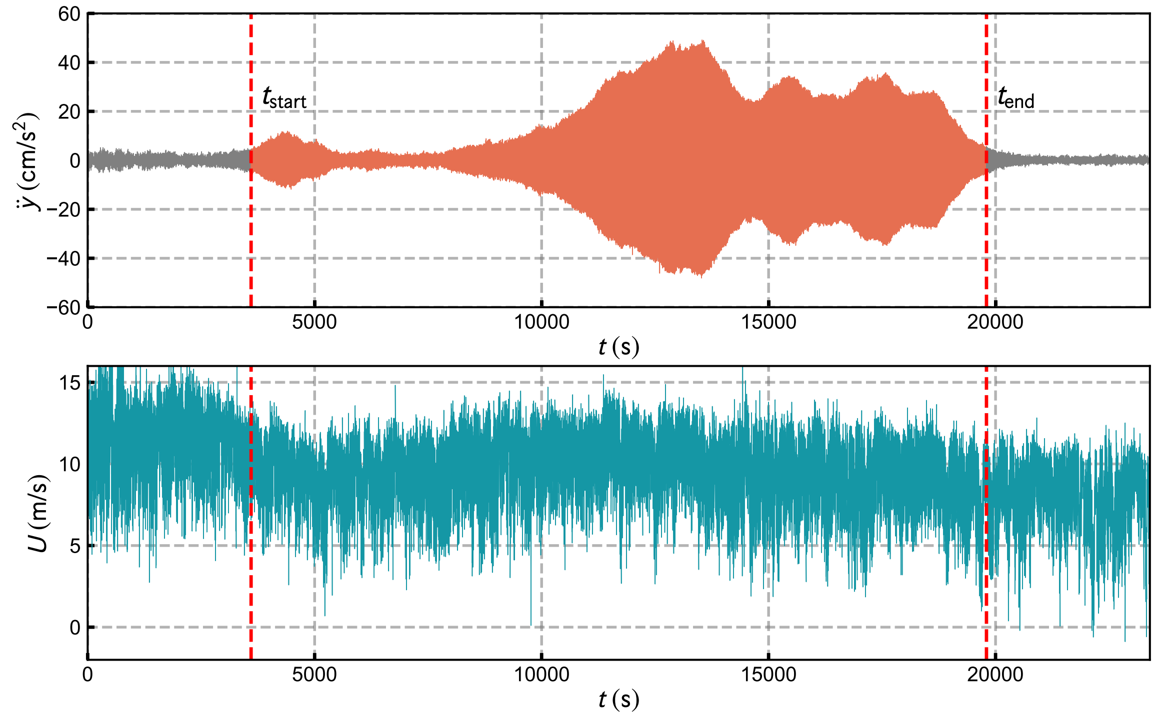
The blue points in Fig. 8 represent the random vibration under the combined action of vehicle and wind field. At high wind speed, wind-induced buffeting takes the main part, and the RMS of the response is relatively high. The points of other colors represent the VIVs of different frequencies, respectively. It can be found that the VIV of each mode has an ambiguous range of wind speed. As the bridge's natural frequency increases, so does the crucial VIV wind speed. Besides, the lock-in ranges of some adjacent VIV modes have apparent overlaps, such as the 0.276 Hz and 0.327 Hz VIV. This phenomenon will be explained in detail later.

Aside from the wind speed and the yaw angle, the other important factors of VIV is the along-wind turbulence intensity I_u, the angle of attack (AOA) α, and the inhomogeneity of wind field along the span-wise direction [9]. As shown in Fig. 9 and Fig. 10, under the southeast wind, the occurrence of the 0.327 Hz VIV is intensely dependent on these wind parameters at the 1/4 span and 3/4 span. Due to the installation mistake of the anemometers, the value of AOA obtained directly is abnormal. The AOA employed here is actually the nominal AOA. When the wind direction is perpendicular to the longitudinal bridge axis and the 10 min average wind speed falls within a certain range, the VIV events may happen, and they are more sensitive to low turbulence condition. These qualitative conclusions are generally consistent with the results of wind tunnel tests. Besides, due to the interference effect of the surrounding mountainous terrain on wind field, the southeast inflow around the 3/4 span has a lower mean wind speed, the considerably stronger turbulence intensity and more diversity in AOA, therefore the wind sensitive to VIV may occur in a broader range of wind speed and turbulence intensity.

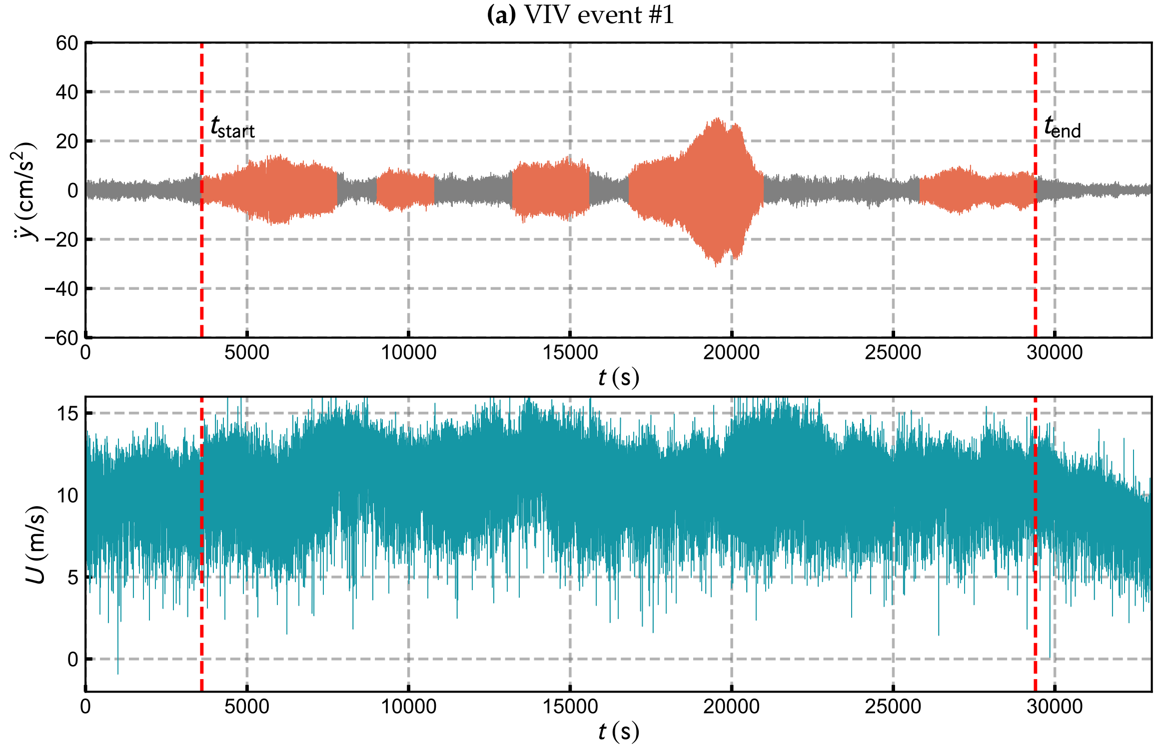
To consider the non-VIV vibration samples from a random perspective, the scatter plots should be transformed into the expression of conditional probability. For example, Fig. 11 give the conditional probability of the 0.327 Hz VIV with respect to wind environment parameters at the 1/4 span. A_{VIV} is the set of events that mean the occurrence of VIV. The conditional probabilities generally show a bell-shaped curve. When the wind parameters are distant from the centers of the distributions, the conditional probabilities gradually reduce to zero. In addition, even at its highest, the VIV conditional probability is much smaller than 1, making it challenging to forecast the VIV of the prototype bridges just based on the wind environment.

The same process is used for the VIV induced by the northwest wind. Ultimately, the basic VIV wind conditions of different modes are summarized in Table 3. n_0 denotes the number of 10 min samples that satisfy the listed VIV conditions, whether they are the VIV or not. n_{VIV} means the number of 10 min VIV samples under fixed mean speed, drift angle and turbulence intensity. (There are some samples that do not meet the wind conditions in Table 3 still have VIV phenomenon, but they are not included in this work due to their tiny quantity (less than 5%). The VIV occurrence probability of the bridge deck under $\boldsymbol{W}$ is measured by the conditional probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W})$, where $\boldsymbol{W}$ represents the given wind environments prone to VIV. Obviously, for different VIV modes, although their number of VIV sensitive samples are analogous, the VIV probability $P(A_{\mathrm{VIV}}|\boldsymbol{W})$ is clearly different. The 0.327 Hz VIV mode is much prone to VIV than that of 0.183 Hz and 0.276 Hz. It should also be noted that the northwest wind $\boldsymbol{W}^{\mathrm{NW}}$ is more likely to cause VIV than the southeast wind $\boldsymbol{W}^{\mathrm{SE}}$, especially for low-frequency modes. A possible explanation for this issue could be that the southeast flow is disturbed by the hills and in front of the 3/4 span, which may weaken the correlation of the vortex street in the spanwise direction. The northwest wind mainly comes from the wide sea surface, and the flow field is more uniform and easier to form a large-scale vortex structure. Regional geometry assures important responsibility for the occurrence of VIV.

The conditional probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}_0)$ is further decomposed under different wind parameters. In Fig. 12(a), as the wind direction



(a) VIV event #1



(b) VIV event #2

Fig. 7. Two types of complete VIV event and the corresponding orthogonal component of the inflow velocity.

deviates from the perpendicular direction, the VIV probability steadily diminishes. The VIV mode is also closely related to the wind direction. The probability of 0.183 Hz VIV is significantly higher in the northwest incoming flow than in the southeast incoming flow. This could indicate that the non-uniformity of the span direction is likely to have a greater impact on the low-frequency mode. The turbulence intensity I_u is also a key factor controlling VIV. As shown in Fig. 12(b), when I_u is less than 0.04, the VIV probability is much higher than that of other conditions. The influence of AOA on the VIV probability is relatively weak. In a wide range ($\alpha \in [11°, 14°]$), the VIV probability $\mathrm{P}(A_{\mathrm{VIV}}|U, \theta, I_u, \alpha)$ considering AOA is approximately equal to the VIV probability $\mathrm{P}(A_{\mathrm{VIV}}|U, \theta, I_u)$ which ignores the AOA. There will be no apparent difference between $\mathrm{P}(A_{\mathrm{VIV}}|U, \theta, I_u)$ and $\mathrm{P}(A_{\mathrm{VIV}}|U, \theta, I_u, \alpha)$ until the AOA is outside the normal range.

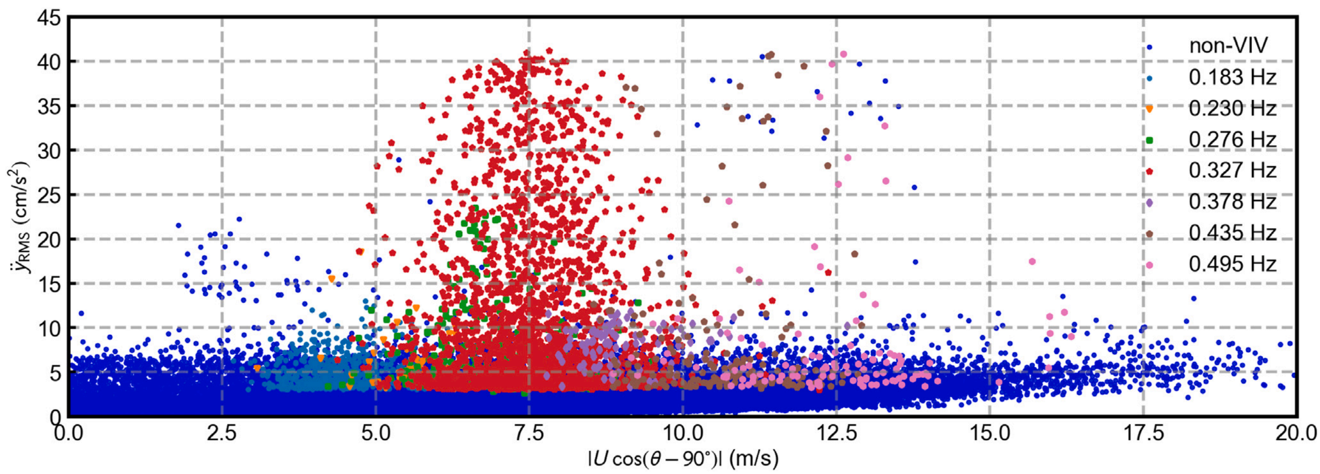


Fig. 8. Variation of the maximum RMS and mode with the wind speed.

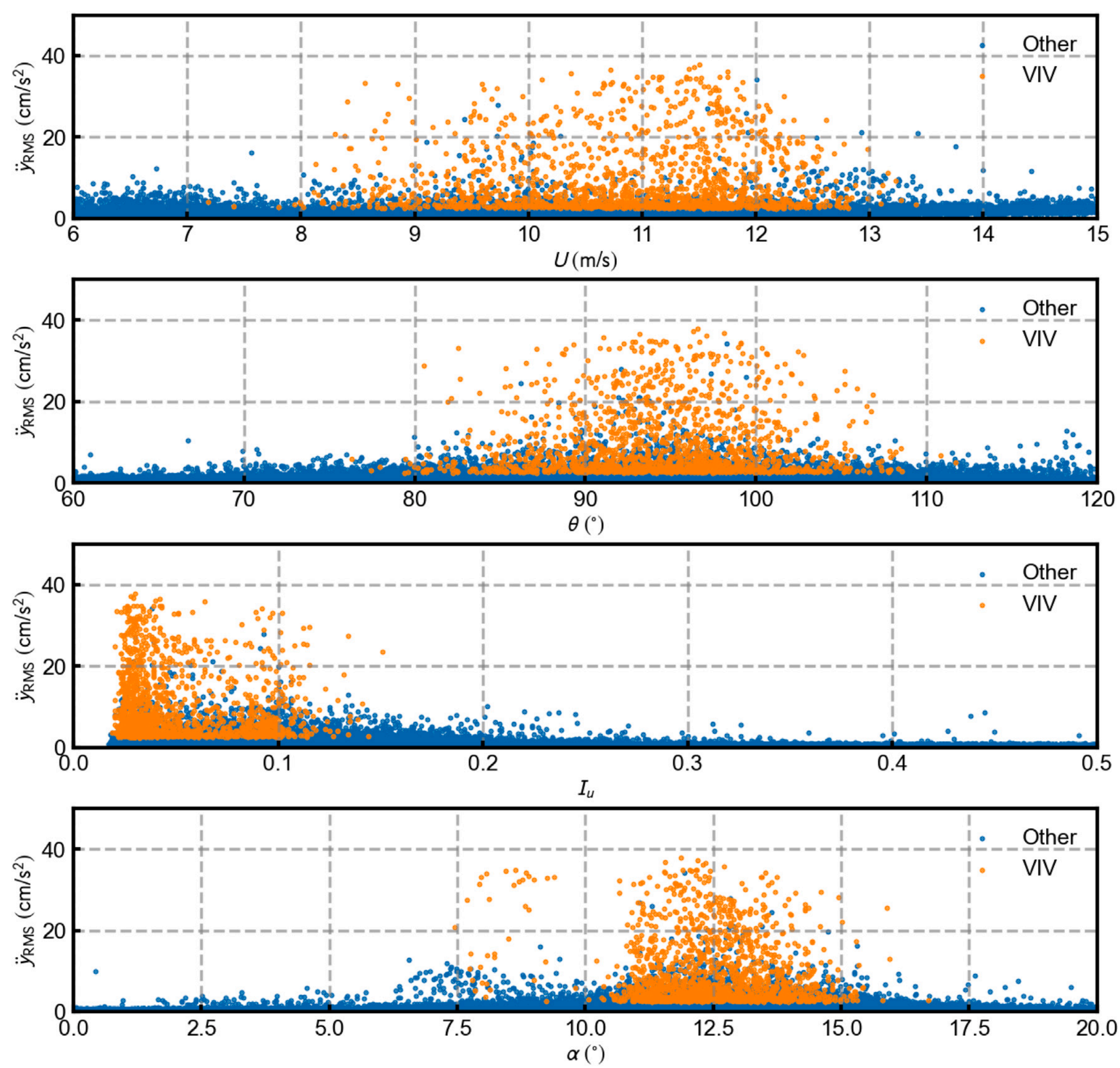


Fig. 9. Distribution of VIV samples of 0.327 Hz with respect to U, θ, I_u and α (1/4 center span, southeast inflow).

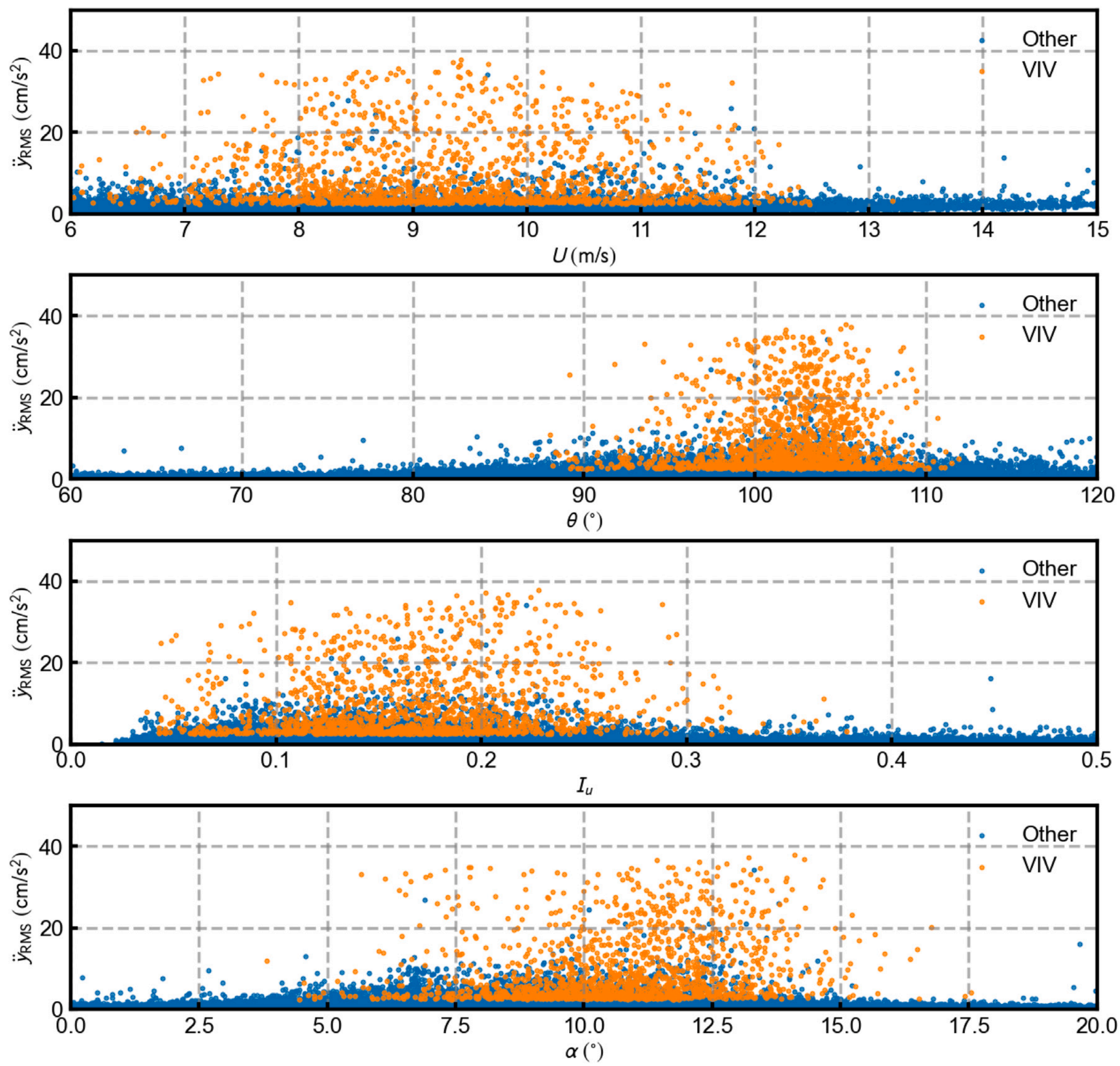


Fig. 10. Distribution of VIV samples of 0.327 Hz with respect to U, θ, I_u and α (3/4 center span, southeast inflow).

3.3. Multi-mode VIV at the same wind speed

After smoothing conditional probability functions by polynomials, as shown in Fig. 13, it is interesting to note the lock-in interval of the 0.276 Hz VIV, especially its high probability part, is almost a subset of that of the 0.327 Hz mode. The red region means that not only the 0.276 Hz VIV may occur in this wind speed range, but also the 0.327 Hz VIV is more likely to occur, which makes it has essential difficulty to accurately determine the VIV mode from the wind speed.

Fig. 14(a) plots a complex VIV event whose eigenfrequency transformed from 0.327 Hz to 0.276 Hz with time. The time–frequency spectrum and time-varying average wind speed of the given VIV event is displayed in the same time coordinate as Fig. 14(a), Fig. 14(b) and Fig. 14(c), respectively. The short-time Fourier transform (SFFT) is used to calculate the time–frequency spectrum, and empirical mode decomposition (EMD) is used to extract the time-varying mean wind. When the VIV event first appeared, the frequency component of 0.327 Hz gained importance and took center stage with the passage of time. Up until around 14,000 s (red imaginary line in Fig. 14), there was no obvious dominating frequency as the 0.276 Hz and 0.327 Hz modes started to compete with each other. Finally, the 0.276 Hz mode outperformed the 0.327 Hz mode, indicating a distinct shift in the VIV pattern. Thus, it is evident from Fig. 14(a) that the investigated bridge's vibration mode changed from symmetric vertical bending to antisymmetric vertical bending. Even if the average wind speed at the measured point did not change clearly, the VIV mode still had the chance to transform. Differently, since the lock-in interval of the 0.183 Hz VIV is far away from the 0.276 Hz and 0.327 Hz VIV (purple region in Fig. 13), and the probability of the 0.230 Hz VIV is relatively low, there is almost no modal transformation of 0.183 Hz VIV with other modes in the whole health monitoring data. It may be considered that the 0.183 Hz is an isolated VIV mode.

4. Relationship between 10 min VIV samples and latent factors

Since the randomness of VIV cannot be fully explained only from the perspective of the observed wind environmental factors, it is necessary to explore the relationship between the occurrence of VIV and unobserved factors. On the one hand, unobserved variables refer to incomplete information of actual flow field, which could not be fully reconstructed by sparsely distributed anemometers due to cost constraints and inevitable sampling errors. Only the wind parameters at the measuring points can be acquired, the distribution and variation of parameters along the span is ambiguous and unpredictable. The other meaning of the unobserved factors is more important. It is often related to the non-wind parameters such as damping ratio and traffic load which are extremely challenging to be measured directly from the BHMS. These non-wind parameters typically have a large degree

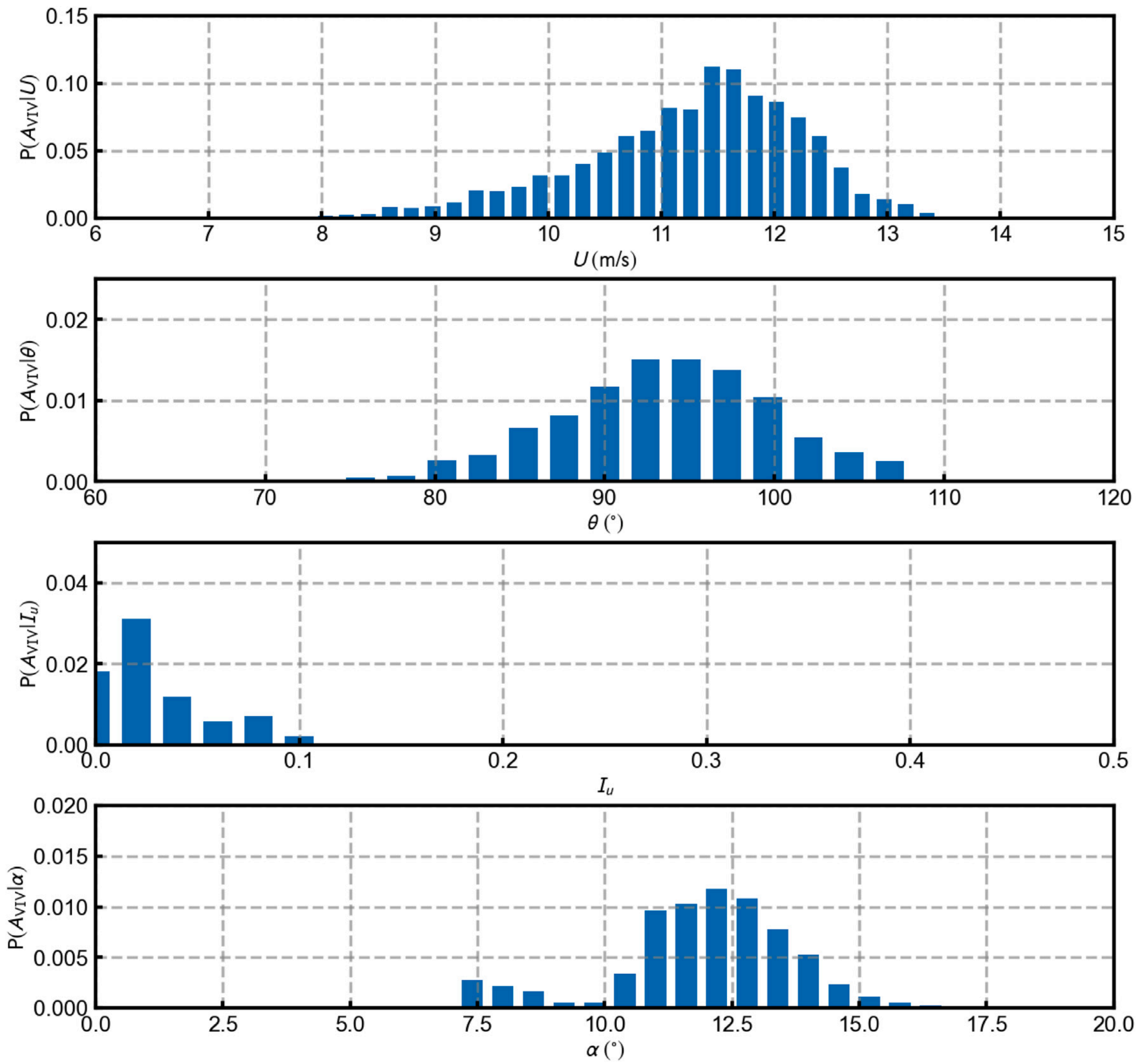


Fig. 11. VIV conditional probability of 0.327 Hz with respect to U, θ, I_u and α (1/4 center span, southeast inflow).

of variability due to aleatoric and epistemic uncertainty. Nonetheless, the impact of non-wind parameters on VIV is important and has not been carefully considered.

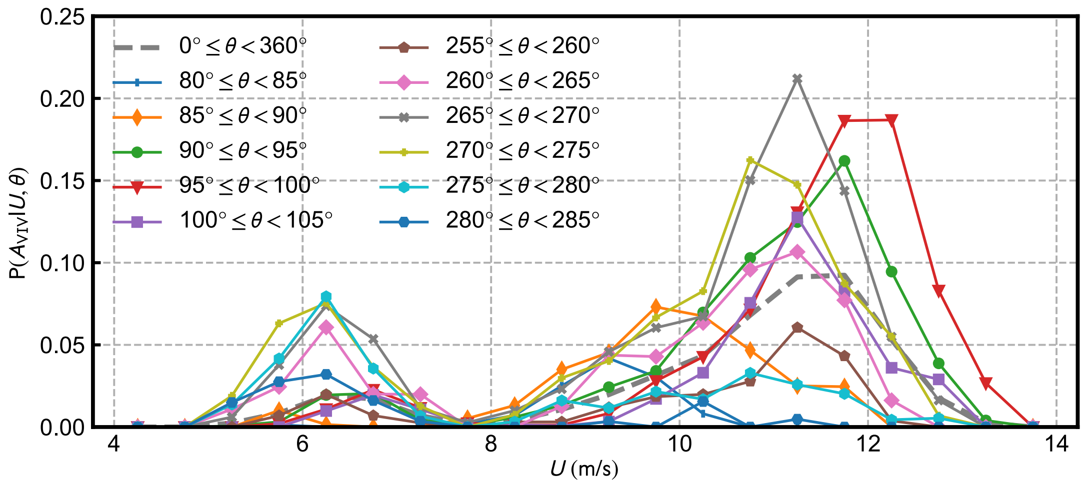
t_h and t_m, which are the hour and month of the occurrence of VIV in one day and one year, respectively, are discussed in this section. The conditional probabilities $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_h)$, $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_m)$ with respect to the fixed VIV sensitive wind condition $\boldsymbol{W}$ are calculated, and the discussion in this section is mainly limited to 0.327 Hz and 0.183 Hz, which are the most easily triggered VIV mode.

4.1. Marginal probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_h)$

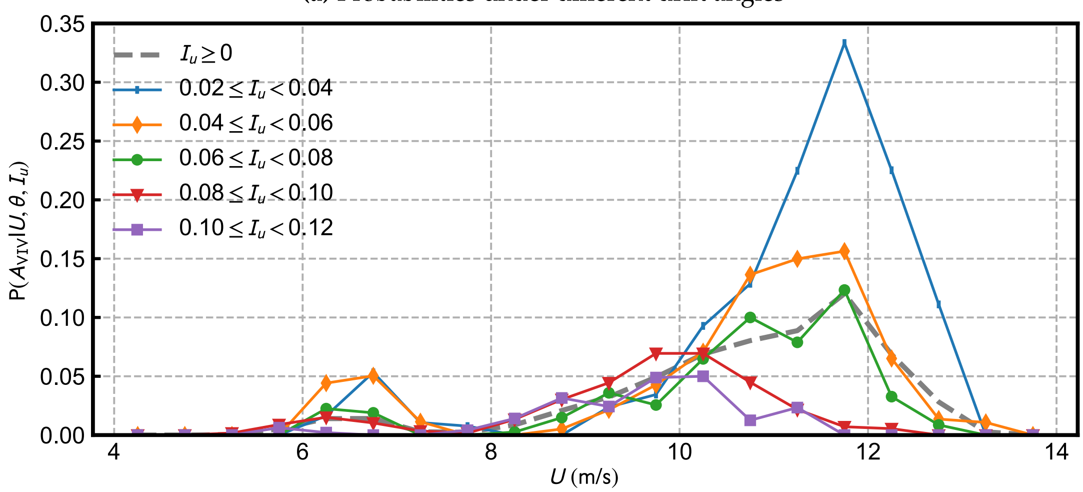
The random variable t_h usually has a strong correlation with the traffic volume per hour. Dan et al. [24] proposed that traffic load leads to the decrease of modal frequency and the increase of modal damping ratio, which can suppress VIV responses. Besides, irregular vehicle queues would significantly change the overall aerodynamic configuration of bridge deck.

For the VIV of 0.183 Hz, Fig. 15 depicts the RMS of acceleration, number and conditional probability of 10 min VIV and non-VIV samples. Some samples with large RMS of acceleration but not be identified as VIV may be wind-induced buffeting. The VIV wind conditions induced by the northwest flow are more evenly distributed over time, while that caused by the southeast flow are mainly concentrated around 12:00 noon. Obviously, there is a noticeable decrease in $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_h)$ near 6:00 and 18:00 when the amount of traffic is heavy because of daily commuting behaviors, indicating a substantial traffic dependence for the VIV. However, for the VIV of 0.327 Hz, as shown in Fig. 16, the VIVs caused by the southeast flow and the northwest flow show completely different results. For the southeast flow, the VIV samples occur frequently at around 18:00, and the distribution of conditional probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_h)$ and VIV sensitive wind is almost consistent at different hours. With the increase of the sample belonging to the set $\boldsymbol{W}$, the VIV probability also increases, and it seems have weak relationship with traffic volume. On the other hand, for northwest inflow, the VIV of 0.327 Hz has a similar law to the VIV of 0.183 Hz, and two troughs can be clearly observed at 6:00 and 18:00 in the graph of conditional probability.

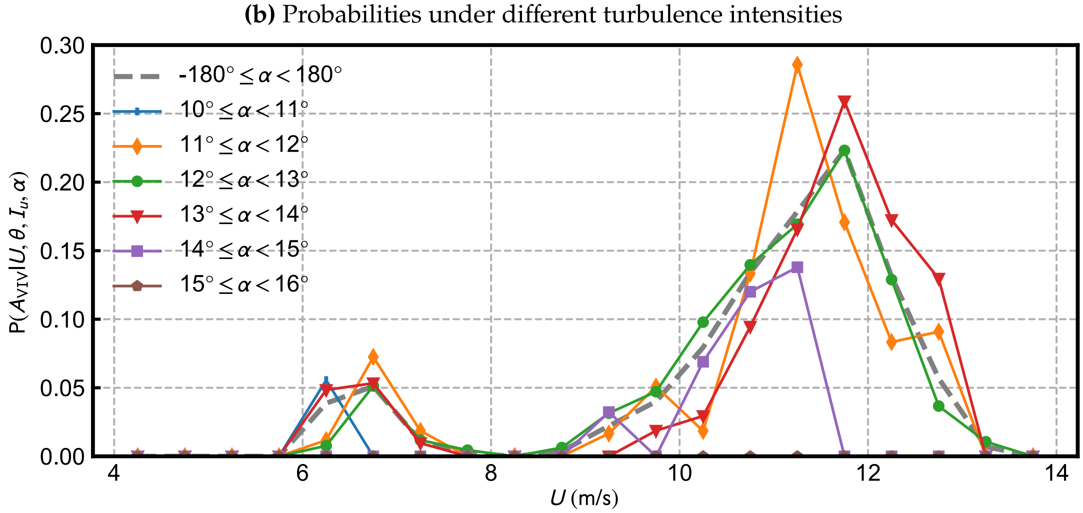
In general, although heavy traffic can effectively suppress the occurrence of the VIV of different modes, especially VIV at low wind speeds, there are still other unknown dominant factors. As shown in Fig. 15(a), the conditional probability is small from 3:00 to 6:00, and its change rule is not completely consistent with the traffic volume. Coincidentally, in Fig. 16(a), the conditional probability seems to be completely determined by the number of VIV-sensitive winds. At a given hour, the more easily the VIV-sensitive wind appears, the higher the conditional probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_h)$ is. Although the traffic volume is large at 18:00, the conditional probability also peaks at this time, and the



(a) Probabilities under different drift angles



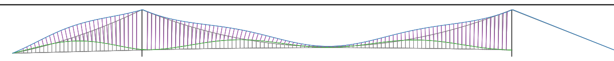
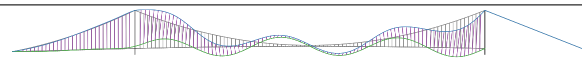
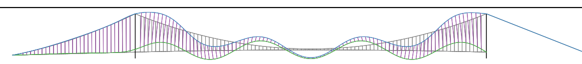
(b) Probabilities under different turbulence intensities



(c) Probabilities under different AOAs

Fig. 12. Different VIV conditional probabilities with respect to wind parameters ((a). Probability $P(A_{VIV}|U,\theta)$; (b). Probability $P(A_{VIV}|U,\theta,I_u)$ where $\theta \in [85°, 95°]$); (c). Probability $P(A_{VIV}|U,\theta,I_u,\alpha)$ where $\theta \in [85°, 95°]$ and $I_u \in [0, 0.06]$)

Table 3
Conclusions of basic VIV wind conditions.

Parameter	Southeast wind		Northwest wind	
	1/4 span	3/4 span	1/4 span	3/4 span
$f = 0.183$ Hz				
Mean speed U (m/s)	[5.4,7.8]	[4.4,7.6]	[5,7.4]	[4,7.4]
Drift angle θ (°)	[84,108]	[90,114]	[252,284]	[260,300]
Turbulence intensity I_u	[0,0.12]	[0,0.28]	[0,0.12]	[0,0.20]
AOA α (°)	[8,15.5]	[4,14]	[6,13]	[2.5,15]
Wind climate samples n_0	21364		8208	
VIV samples n_{VIV}	218		271	
VIV conditional probability	0.010		0.033	
$f = 0.276$ Hz				
Mean speed U (m/s)	[7.5,10.4]	[6.6,10.2]	[8,11.5]	[7,10]
Drift angle θ (°)	[84,108]	[92,110]	[255,279]	[268,296]
Turbulence intensity I_u	[0,0.12]	[0,0.30]	[0,0.10]	[0,0.18]
AOA α (°)	[7.5,16]	[5,13.5]	[5,11]	[5,12.5]
Wind climate samples n_0	20107		5359	
VIV samples n_{VIV}	93		141	
VIV conditional probability	0.005		0.026	
$f = 0.327$ Hz				
Mean speed U (m/s)	[8,13.2]	[6.8,12.4]	[9.4,12.6]	[8,13.2]
Drift angle θ (°)	[78,108]	[88,112]	[254,278]	[266,298]
Turbulence intensity I_u	[0,0.12]	[0,0.36]	[0,0.12]	[0,0.20]
AOA α (°)	[7.5,16]	[5,16]	[7.5,14]	[2.5,15]
Wind climate samples n_0	31655		8154	
VIV samples n_{VIV}	1418		545	
VIV conditional probability	0.045		0.067	

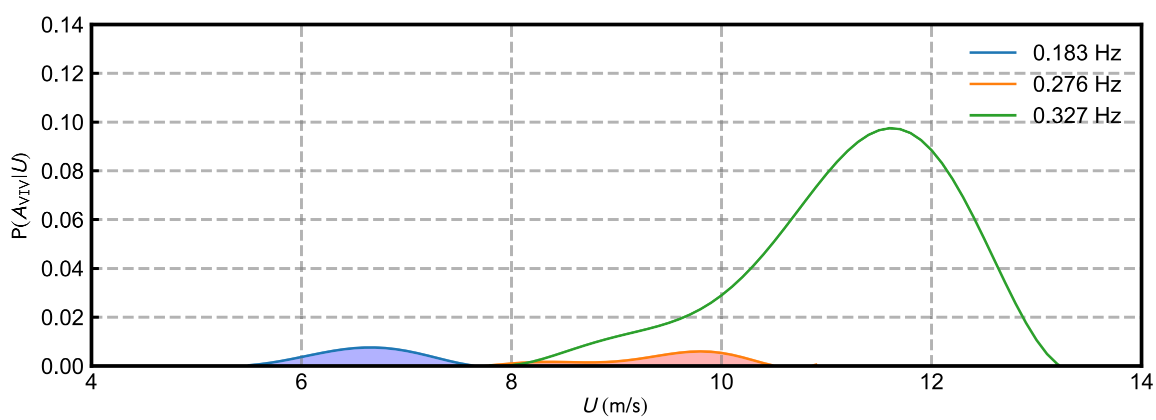


Fig. 13. Conditional probability of different VIV modes (1/4 center span, southeast inflow).

inhibitory effect of traffic load would be ignored. Therefore, the general rules applicable to various modes and wind directions have not yet been found.

4.2. Marginal probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_{\mathrm{m}})$

Different from t_{h}, t_{m} is usually related to the long-period ambient temperature, which may affect the damping ratio of the structure to a certain extent and increase the risk of VIV. In addition, t_{m} represents some implicit information of large-scale wind field structure. As shown in Figs. 18 and 19, in most months, the southeast inflow is less likely to cause VIV than the northwest inflow for both 0.183 Hz VIV and 0.327 Hz VIV. Fig. 17 plots the change in the wind rose graph over 2013 at two-month intervals, and the joint distribution of wind speed and direction for other years is broadly similar to that of 2013. From January to April, the southeast wind continues to strengthen, a large number of wind conditions that can cause 0.183 Hz VIV or 0.327 Hz VIV can be observed, but the VIV probability $P(A_{\mathrm{VIV}}|\boldsymbol{W}, t_{\mathrm{m}})$ induced by southeast wind has no significant change. In the northwest direction, the number of high wind speed samples decreases rapidly, and the number of low wind speed samples does not decrease much. Although the probability of northwest wind is getting lower and lower with the month, but the VIV probability $P(A_{\mathrm{VIV}}|\boldsymbol{W}, t_{\mathrm{m}})$ continuously increases. In May and June, the western Pacific subtropical high begins to form the local climate in eastern China. A wide-ranging southeast monsoon should be the predominant wind environment. Currently, the northwest and southeast strong winds are somewhat diminished, with the southeast low wind taking center stage. VIVs of different wind directions and modes are all relatively easy to arise, especially when the incoming flow is orthogonal northwest strong wind. Until July and August, the wind field tends to be stable and the temperature reaches the maximum value in a year. The VIV samples are mainly generated by the southeast wind, and the conditional probability is generally close to that of the previous two months. The summer–winter monsoon transition happens

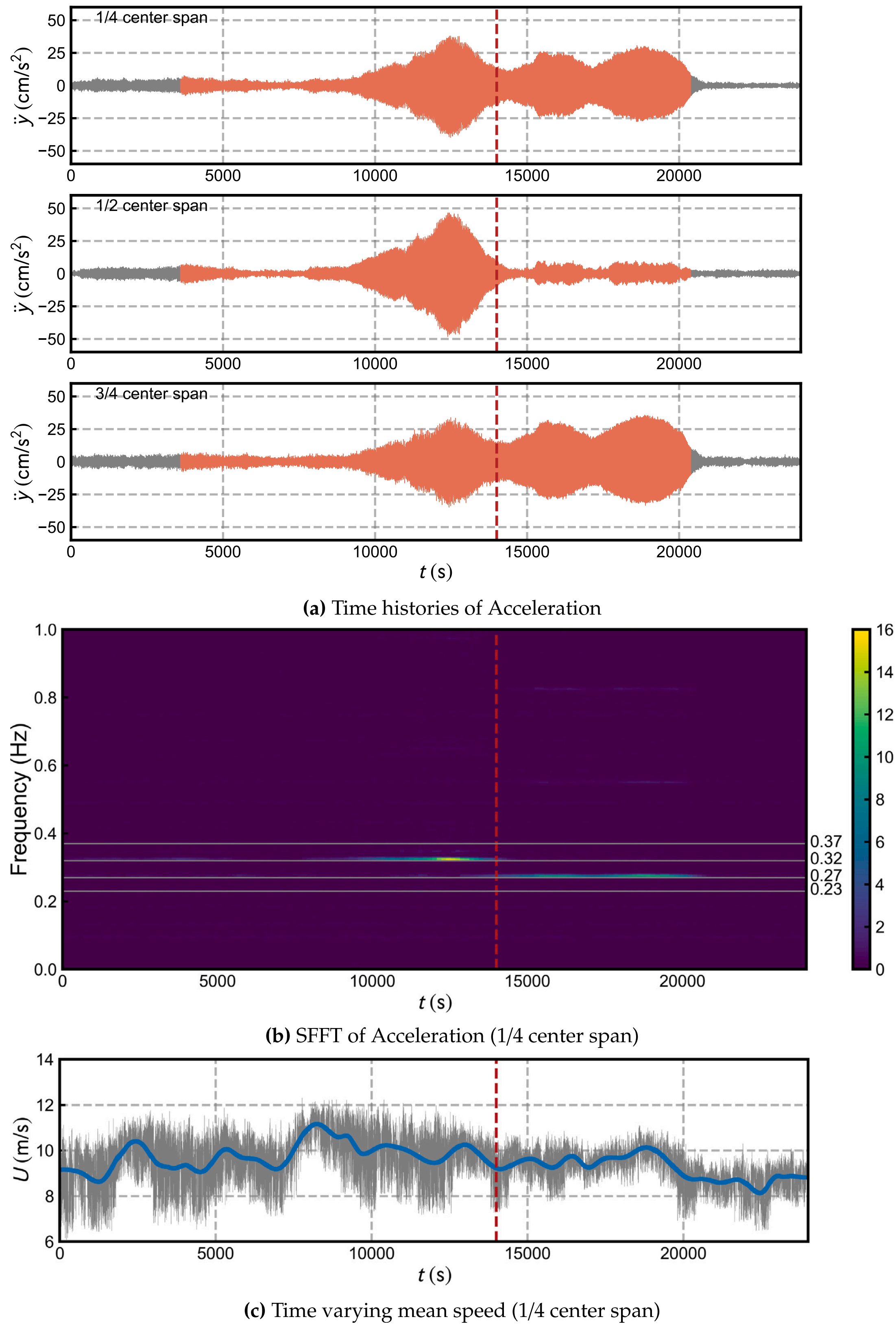
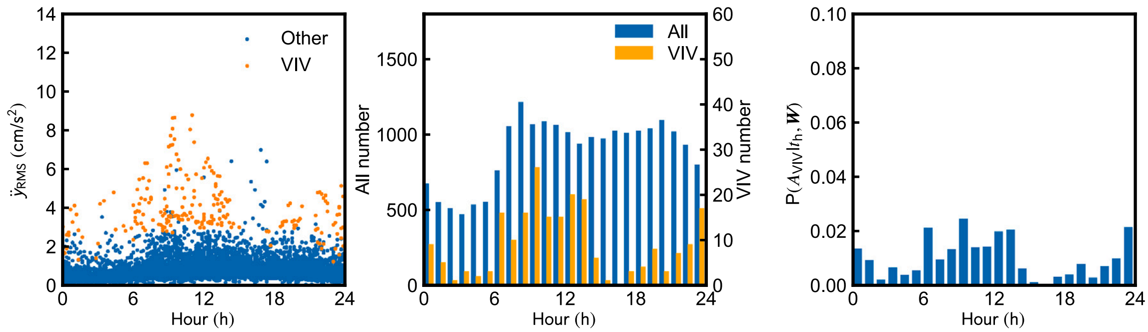
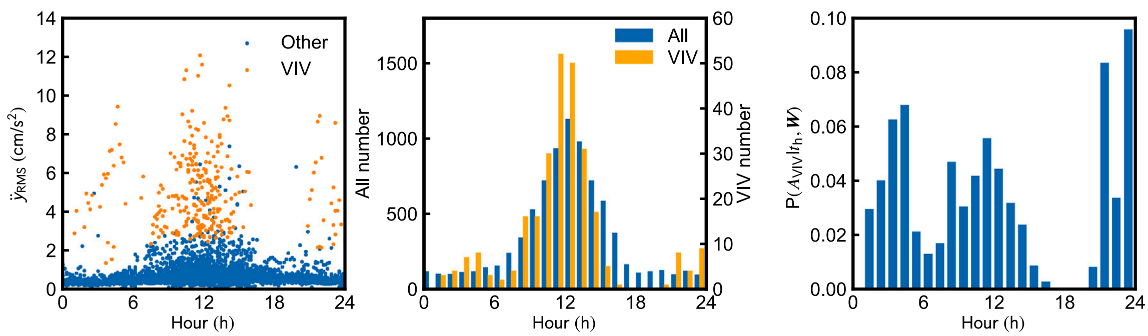


Fig. 14. Complex VIV event from 0.327 Hz to 0.276 Hz.

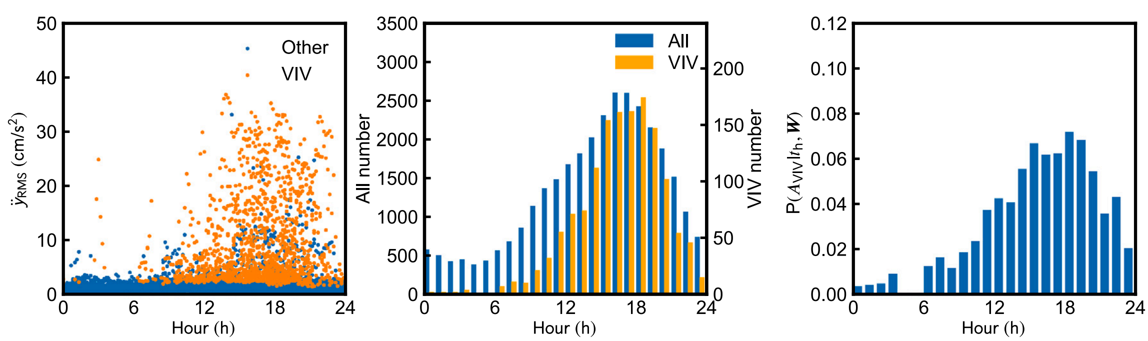


(a) Southeast wind

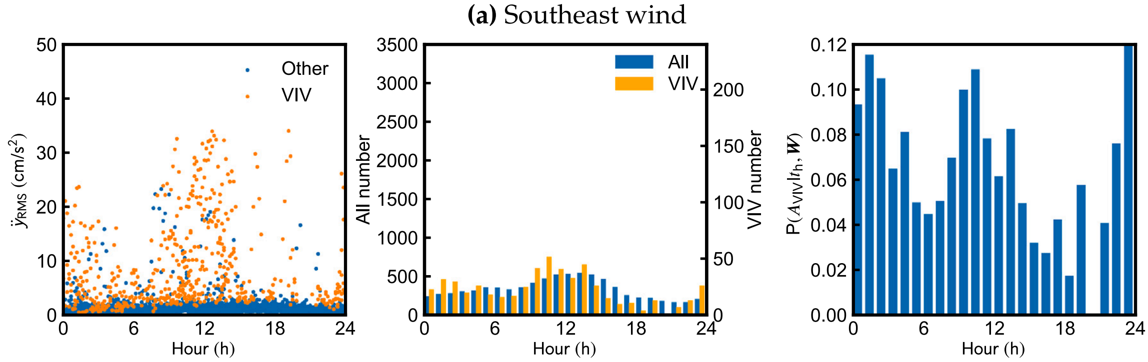


(b) Northwest wind

Fig. 15. Relationship between 0.183 Hz VIV and hourly period.



(a) Southeast wind



(b) Northwest wind

Fig. 16. Relationship between 0.327 Hz VIV and hourly period.

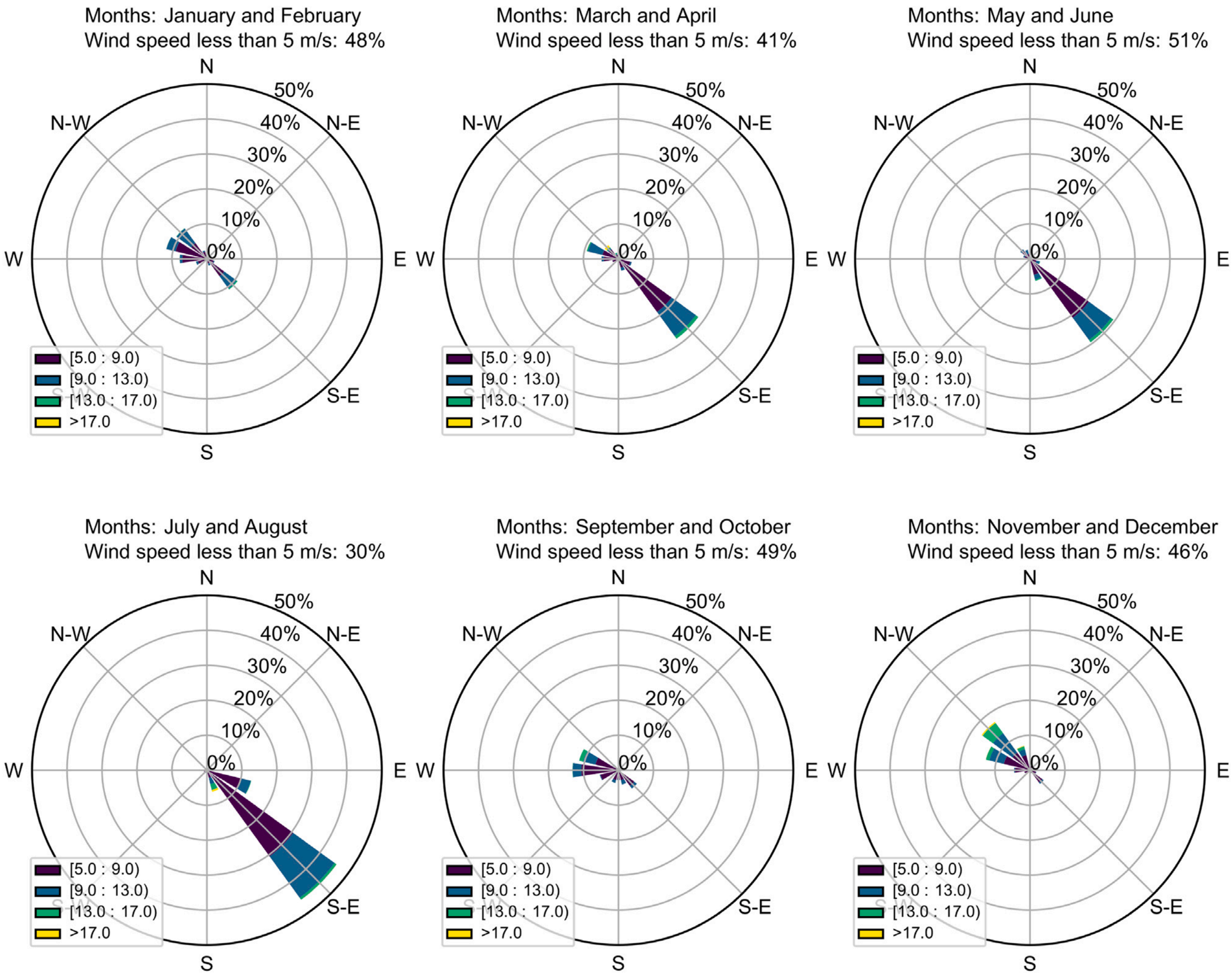


Fig. 17. Wind rose charts for different months in 2013 (unit:m/s) .

from September to October, once winter arrives, the VIV sensitive wind usually originates in the northwest.

It is evident that there is a significant seasonal dependence on the VIV probability. During the cold winter months, from November to January, despite the presence of a large number of northwest monsoon winds blowing from the continent to the ocean, a few VIV samples are actually observed during this time period. In summer, from July to August, northwest inflows sensitive to VIV are quite sparse, especially for the 0.327 Hz VIV, but the conditional probability is still high. VIV appear to be more prevalent in the summer and autumn but relatively uncommon in the winter. This phenomenon would be partially attributed to the yearly seasonality of damping ratio. Chu et al. [25] used the Bayesian FFT identification algorithm to decompose the time series of damping ratios and then proposed that the yearly fluctuating ambient interference might influence the damping ratio and the increment of temperature can decrease the damping ratio. Six vertical accelerometers are used to extract the dynamic characteristics (AC10, AC12, AC13, AC15, AC16 and AC18 in Fig. 4). More details about the Bayesian FFT identification algorithm are shown in [25]. Fig. 20 plots the monthly average damping ratios of the fourth-order symmetric vertical mode (0.327 Hz). In May and June, the damping ratio is comparatively small, whereas the VIV conditional probability $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_{\mathrm{m}})$ is high. The damping ratio reaches its annual maximum in December, and $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_{\mathrm{m}})$ is almost zero. It is not until the damping ratio begins to decrease in January that VIV occurs frequently. As shown in Fig. 21, the conditional probability and damping ratio show a significant negative correlation, and the VIV is almost impossible to occur when the damping ratio reaches 4‰. Compared with the VIV induced by southeast wind, the VIV induced by northwest wind is more sensitive to the damping ratio varying with season, and the conditional probability decays rapidly from summer to winter. Besides, the fitting relationship can be used to guide the design of TMDs approximately. TMDs should make the damping ratio greater than the critical damping ratio (the damping ratio where the VIV probability is zero, it is 4.05‰in Fig. 21), so that the VIV can be almost completely controlled. Obviously, like t_{h}, the above relationships of conditional probabilities with seasons are only approximate laws, and those non-wind parameters implicit in the statistical results that really determine whether VIV occurs or not need to be further investigated.

As is shown in Fig. 22, the conditional probability can be further decomposed with respective to t_{m}. Comparing with probability without time-varying characteristics (gray line in Fig. 22), the VIV probability $\mathrm{P}(A_{\mathrm{VIV}}|U, \theta, I_u, t_{\mathrm{m}})$ is low in March and November, which matches with the two damping ratio peaks seen in Fig. 22. Since the VIV of the prototype bridge is not only affected by observable wind parameters, but also by latent surrounding factors, it is necessary to consider seasonal characteristics when using statistical inference methods to assess VIV, which means that only a sufficiently large data set can fully reveal the characteristics of VIV. Using data from short-term BHMS to assess the VIV performance may be biased. Short-term BHMS is greatly disturbed and cannot properly reflect the characteristics of annual seasonal fluctuation. As an example, Smith [1] collected 116 days of wind speed and vibration data to predict the future response of the Wye

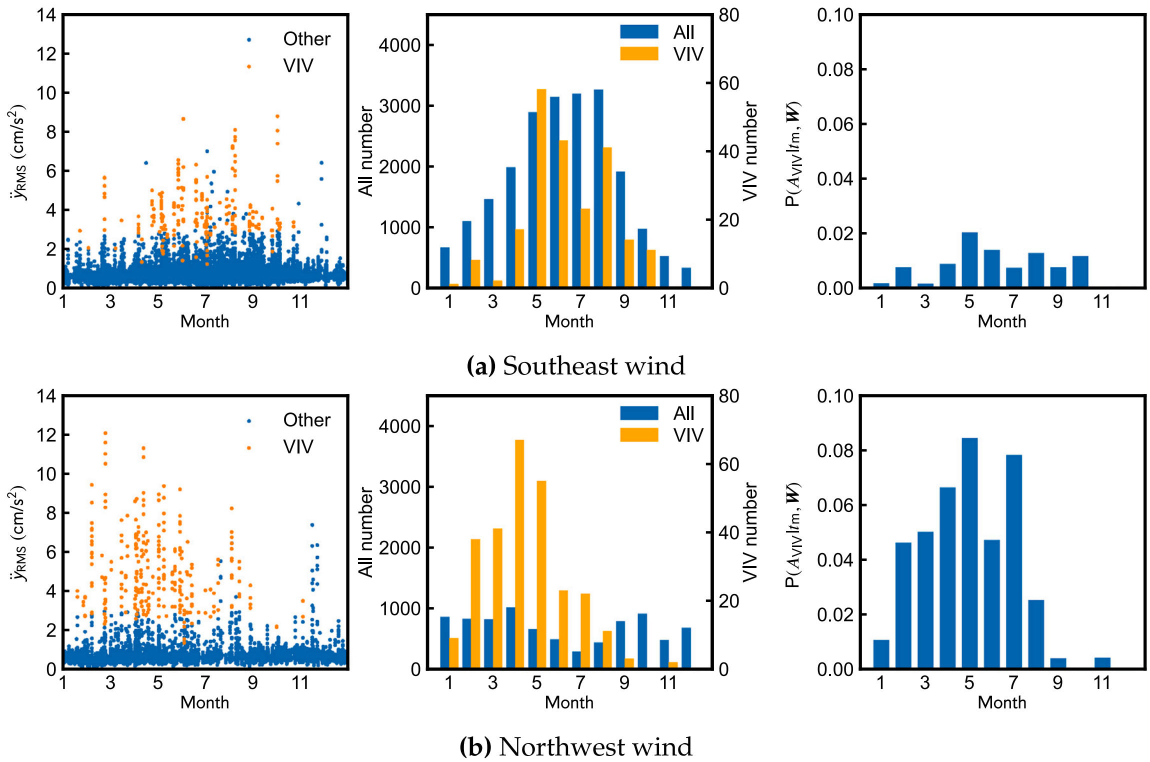


Fig. 18. Relationship between 0.183 Hz VIV and monthly period.

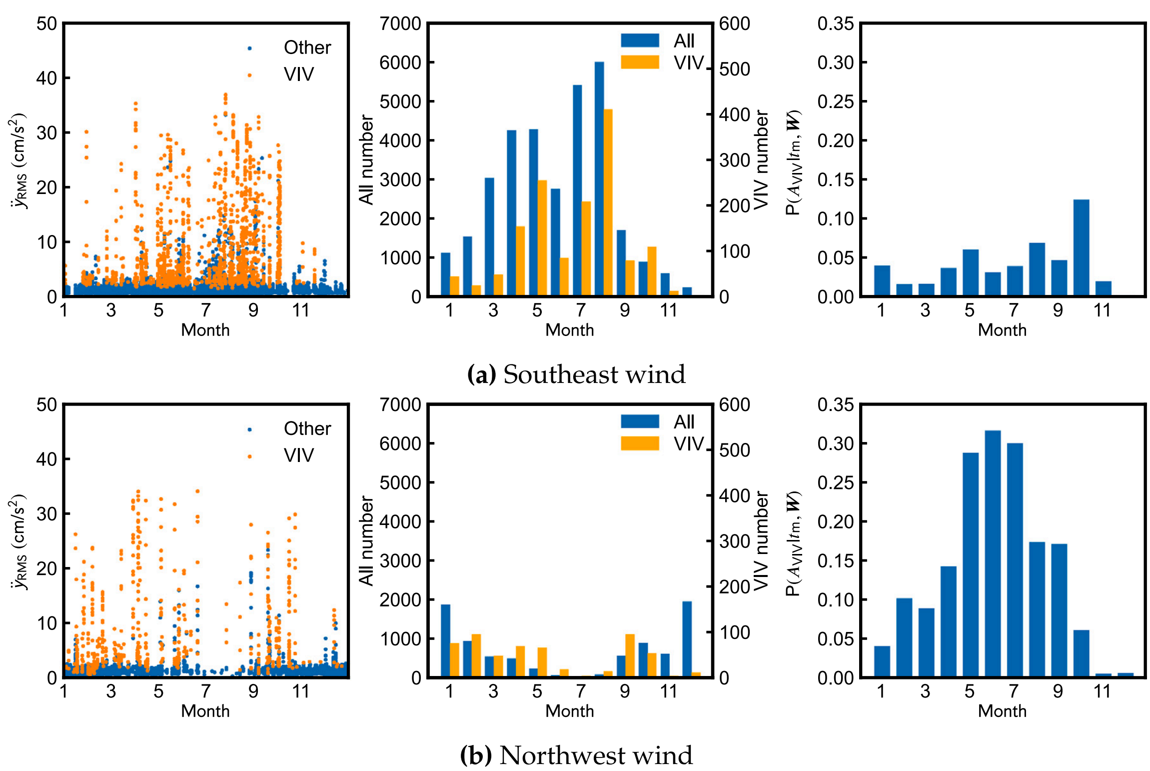


Fig. 19. Relationship between 0.327 Hz VIV and monthly period.

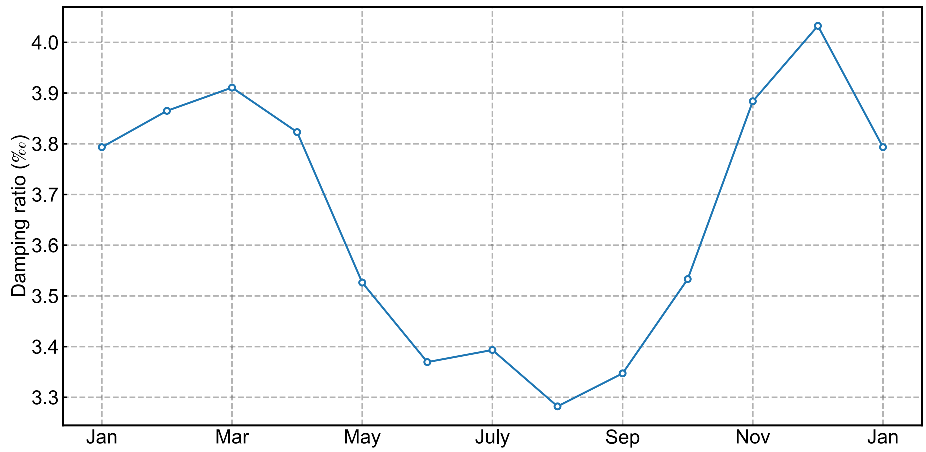


Fig. 20. Monthly average damping ratio of the vibration mode with the frequency of 0.327 Hz.

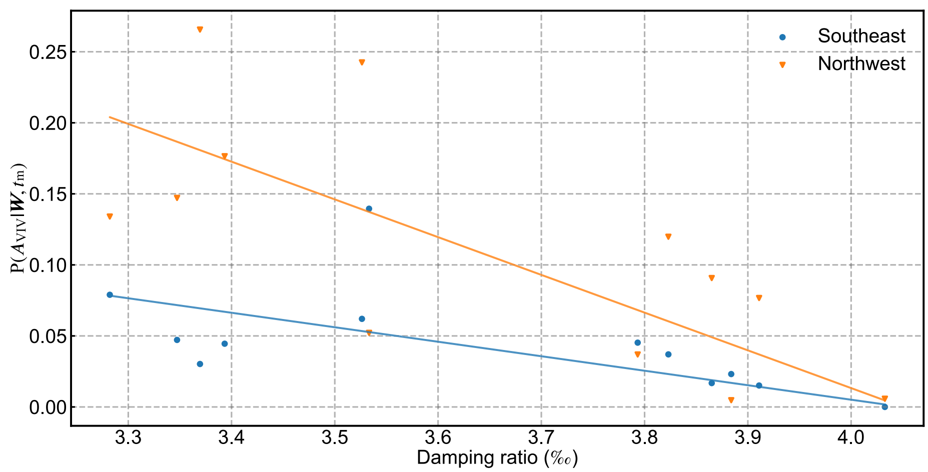


Fig. 21. Linear relationship between damping ratio and $\mathrm{P}(A_{\mathrm{VIV}}|\boldsymbol{W}, t_{\mathrm{m}})$.

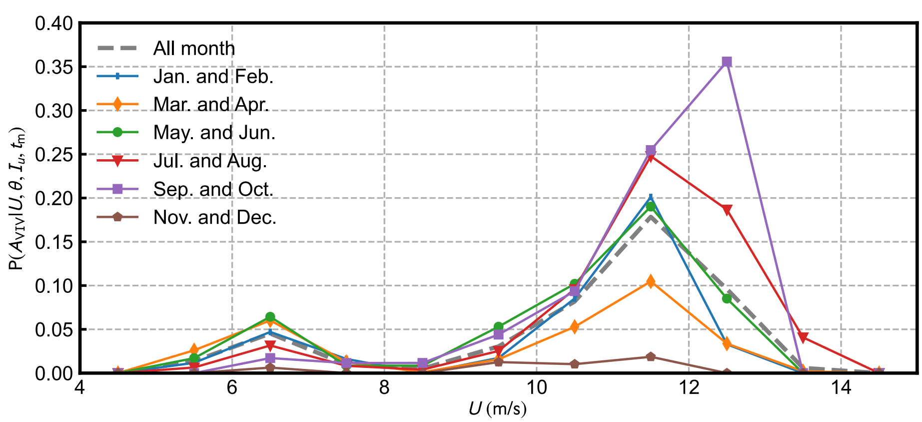
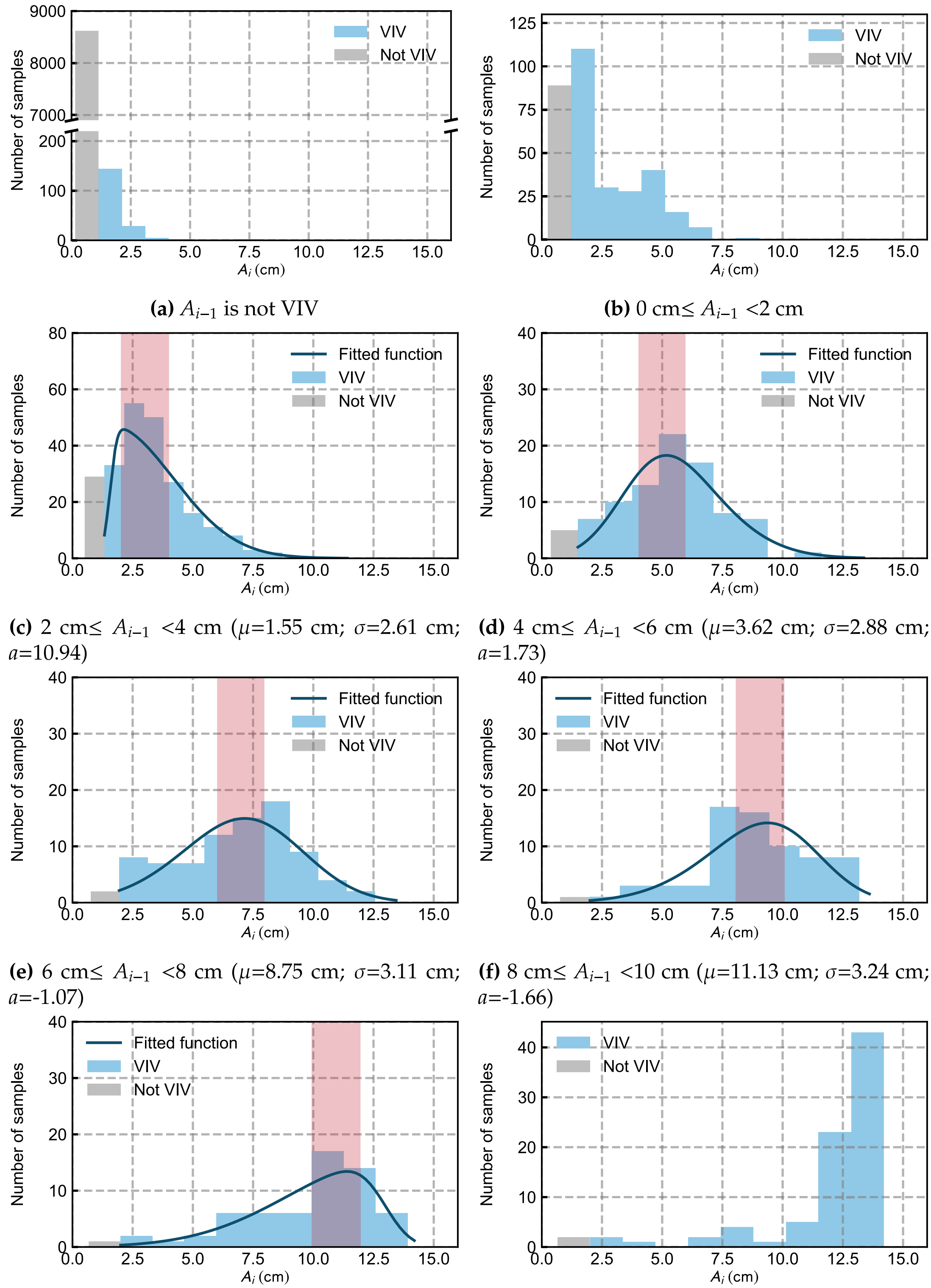


Fig. 22. Different VIV conditional probabilities with respect to months (Probability $\mathrm{P}(A_{\mathrm{VIV}}|U, \theta, I_u, t_{\mathrm{m}})$ where $\theta \in [75°, 105°] \cap [255°, 285°]$ and $I_u \in [0, 0.06]$).

Bridge, and then inferred the quantity of oscillation cycles in one year to assess the impact of VIV on fatigue failure. Nonetheless, the winter and spring when the damping ratio of bridge may be high make up the majority of the collected data, which could be the primary cause of the low VIV conditional probability (0.847%).

5. Time series characteristics of adjacent VIV samples

The previous sections mainly discuss the relationship between discrete 10 min VIV samples and wind or non-wind parameters. This chapter attempts to shed more light on the amplitude evolution of time-adjacent samples. Limited by the available samples, only the 0.327 Hz VIV induced by the southeast wind is discussed here. Fig. 23 shows the variation of A_i under different given A_{i-1}, where A_i is displacement amplitude at the ith step, and it represents the conditional probability $\mathrm{P}(A_i|\boldsymbol{W}_i, A_{i-1})$. The introduction of $\mathrm{P}(A_i|\boldsymbol{W}_i, A_{i-1})$ implies an assumption of Markov property of VIV development, so the amplitude of the current time step is only related to the amplitude of the previous time step and the external information of the current time step. As shown in



(a) A_{i-1} is not VIV

(b) 0 cm≤ A_{i-1} <2 cm

(c) 2 cm≤ A_{i-1} <4 cm (μ=1.55 cm; σ=2.61 cm; a=10.94)

(d) 4 cm≤ A_{i-1} <6 cm (μ=3.62 cm; σ=2.88 cm; a=1.73)

(e) 6 cm≤ A_{i-1} <8 cm (μ=8.75 cm; σ=3.11 cm; a=-1.07)

(f) 8 cm≤ A_{i-1} <10 cm (μ=11.13 cm; σ=3.24 cm; a=-1.66)

(g) 10 cm≤ A_{i-1} <12 cm (μ=12.96 cm; σ=3.97 cm; a=-4.51)

(h) 12 cm≤ A_{i-1}

Fig. 23. Distribution of displacement amplitude A_i under different A_{i-1} (is the range of A_{i-1})

Fig. 23(a), if there is no VIV in the previous step, the bridge is difficult to fall in the VIV state (2%). Once the bridge starts to oscillate, even in a very small amplitude less than 2 cm, the probability of maintaining VIV will be significantly higher than before (80%). Moreover, this state with small displacement amplitude is still relatively unstable because there still exist probability about 20% to escape VIV. As the displacement of the previous time step further increases, from Fig. 23(c) to Fig. 23(g), it shows that a few VIV samples decay although the wind condition is benefit to VIV, and the development trajectories are not completely consistent with the classical VIV theory. The amplitude distribution gradually shows a skewed Gaussian distribution. The PDF satisfies

$$f(A_i = x|A_{i-1}, \boldsymbol{W}_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{(x-\mu)^2}{2\sigma^2}\right) 2\Phi\left(a\frac{x-\mu}{\sigma}\right) \tag{3}$$

where Φ is the cumulative distribution function of standard normal distribution, a here is called the shape parameter which modulates the skewness of PDF. From Fig. 23(c) to Fig. 23(g), μ is less than A_{i-1} when A_{i-1} is small; Conversely, for larger A_{i-1}, μ will become larger than A_{i-1}. The corresponding shape parameter a changes from positive to negative and there is a slight increase in σ. It is indicated that the development of VIV in large amplitude is less likely to be suppressed by random disturbances comparing with that in small amplitude. Once the higher VIV amplitude (Fig. 23(h)) is reached, most samples can also maintain the current state easily, so fully developed VIV seems hardly to be weakened by random disturbances such as traffic flow and spatial inhomogeneity.

As an ideal way of modeling, the displacement amplitude $\{A_i\}$ at adjacent 10-min is assumed to be conditionally Markovian by discretizing the vibration state of the bridge, while wind parameters $\{\boldsymbol{W}_i\}$ functions as an exogenous variable. The assumptions state that:

$$\begin{aligned}
&\mathrm{P}(\boldsymbol{W}_s|\boldsymbol{W}_{s-1}, A_{s-1}, \boldsymbol{W}_{s-2}, A_{s-2}, \ldots, \boldsymbol{W}_0, A_0) = \mathrm{P}(\boldsymbol{W}_s|\boldsymbol{W}_{s-1}) \\
&\mathrm{P}(A_s|\boldsymbol{W}_s, \boldsymbol{W}_{s-1}, A_{s-1}, \boldsymbol{W}_{s-2}, A_{s-2}, \ldots, \boldsymbol{W}_0, A_0) = \mathrm{P}(A_s|\boldsymbol{W}_s, A_{s-1})
\end{aligned} \tag{4}$$

Therefore, under given wind parameters $\boldsymbol{W}_s, \ldots, \boldsymbol{W}_0$, the conditional probability $\mathrm{P}(A_s|A_0, \boldsymbol{W}_s, \ldots, \boldsymbol{W}_0)$ at the time step s can be decomposed into

$$\begin{aligned}
\mathrm{P}(A_s|A_0, \boldsymbol{W}_s, \ldots, \boldsymbol{W}_0) &= \sum \mathrm{P}(A_s, \ldots, A_1|A_0, \boldsymbol{W}_s, \ldots, \boldsymbol{W}_0) \\
&= \sum \left(\prod_{i=1}^{s} \mathrm{P}(A_i|\boldsymbol{W}_i, A_{i-1})\right)
\end{aligned} \tag{5}$$

in which the sign of summation represents the different values of random variables A_1 to A_{s-1} in the state space. The variation of $\boldsymbol{W}_i$ is not considered here for simplicity, so $\{A_i\}$ becomes a homogeneous Markov chain. The transition matrix $\mathbb{P}$, which represents the evolution pattern of the observed value from the current moment to the next moment, can be utilized to measure the probability of each state at any time in future. Different vibration states are noted as e_0, e_1, e_2, $\ldots$, e_7 here, where e_0, e_i $(1 \le i \le 6)$, e_7 means VIV does not occur, VIV with displacement amplitude from $2i-2$ cm to $2i$ cm and VIV with displacement amplitude greater than 12 cm, respectively. The transition matrix $\mathbb{P}$ is

$$\begin{bmatrix}
0.979 & 0.016 & 0.005 & 0 & 0 & 0 & 0 & 0 \\
0.276 & 0.295 & 0.217 & 0.180 & 0.026 & 0.003 & 0.003 & 0 \\
0.123 & 0.110 & 0.496 & 0.182 & 0.068 & 0.017 & 0.004 & 0 \\
0.056 & 0.034 & 0.135 & 0.393 & 0.258 & 0.101 & 0.012 & 0.011 \\
0.024 & 0.012 & 0.133 & 0.145 & 0.289 & 0.313 & 0.048 & 0.036 \\
0.014 & 0 & 0.029 & 0.057 & 0.214 & 0.4 & 0.157 & 0.129 \\
0.016 & 0 & 0.016 & 0.049 & 0.148 & 0.148 & 0.475 & 0.148 \\
0.024 & 0 & 0 & 0.012 & 0.037 & 0.049 & 0.134 & 0.744
\end{bmatrix} \tag{6}$$

where the row i and column j of the matrix represent the transition probability from e_{i-1} to e_{j-1}. The value of the matrix is estimated from the number of the VIV samples of different amplitudes in Fig. 23. Starting from different initial displacement conditions, the probability that the VIV is greater than 12 cm at different times is given in Fig. 24. When the initial state is the small amplitude or non-VIV, the VIV probability falling in the LCO is increasing. While the initial state is the large amplitude, the VIV probability falling in the LCO increases first and then decreases. This methodology gives a quantitative description of the risk of short-term VIV.

Obviously, there exists $k \in \mathbb{N}$ such that there is no 0 in $\mathbb{P}^k$, the Markov chain is therefore equipped with ergodicity, and the stationary distribution converges to a rank-one matrix with the same row:

$$\begin{bmatrix} 0.768 & 0.025 & 0.039 & 0.034 & 0.035 & 0.034 & 0.025 & 0.039 \end{bmatrix} \tag{7}$$

Table 4
Abilities to trigger VIV in different wind parameter ranges (n_{not}: the number of samples which keeps stable; n_{start}: the number of samples which transform from the stable state to VIV)

Parameter			n_{not}	n_{start}	$n_{\mathrm{start}}/n_{\mathrm{not}}$
Mean speed U (m/s)	Yaw angle θ (°)	Turbulence intensity I_u			%
[9.5,12.5]	[80,100]	(0,0.08]	8620	188	2.18
[9.5,12.5]	[85,95]	(0,0.08]	3904	108	2.77
[10,12]	[80,100]	(0,0.08]	4044	95	2.34
[9.5,12.5]	[80,100]	(0,0.04]	5693	146	2.56
[10,12]	[85,95]	(0,0.04]	610	28	4.59

The long-term behavior of bridge tends not to occur VIV, since the main part of $\mathbb{P}$ is concentrated in $\mathrm{P}(A_i = e_0|A_{i-1} = e_0, \boldsymbol{W})$. As the time step increases, this term gradually spreads to different positions of the matrix. After a long enough time, any initial state will eventually evolve into a non-VIV state. This Markov model is unfortunately not suitable for long-term VIV probability evaluation, because it is still unclear to explain why it is so difficult for the bridge to start VIV. By further compressing $\boldsymbol{W}$ to a smaller range (Table 4), the ratio of the number of samples starting VIV to the number of samples that remain stationary does rise greatly, however, it is still far less than 1. The threshold that prevents the VIV from reaching the germination state continually exists, even in the extremely sensitive wind parameters. It is indicated that more than 10 min of flow field information or instantaneous random disturbance is crucial to the initial stage of VIV. The Markov chain here does use the information more than 10 min to predict VIV, which may destroy the integrity of VIV events in the time domain, since VIV samples actually occur continuously on the time axis. For prototype bridges, the mechanism of the development from the stable state to the VIV state remains to be further studied.

Finally, the difference between the prediction of VIV and other wind-induced vibrations is worth highlighting. Flutter primarily happens at high wind speed. During this period, wind speed is the key factor, and variations in environmental conditions have minimal impact. Buffeting shares similar characteristics, allowing for the satisfactory output by predicting it solely based on wind parameters. For VIV, the amazing stochasticity much stronger than other vibrations has been observed in the field measurement, at least in terms of RMS. Under the same wind condition, whether VIV will occur and the RMS of VIV are very discrete and cannot be deterministically predicted, and the VIV probability is influenced by environmental conditions, including traffic flow and the damping ratio, to a certain degree. More important, the self-excited force in flutter is mainly determined by a linear combination of structural displacement and velocity, the buffeting force contains the linear component of the fluctuating wind speed in different directions, and there are simpler nonlinear patterns in the self-excited force and the buffeting force than in VIV. Conversely, vortex-induced force is primarily dominated by amplitude-dependent characteristics that exhibit a nonlinear relationship with structural displacement and velocity, rather than a linear one, so its prediction is more complicated than other vibrations. VIV may become stronger, weaker or even disappear during the amplitude development period, and the structural amplitude of the previous time step needs to be considered in the VIV prediction. The proposed framework is specifically designed to deal with the main characteristics of VIV, such as high randomness, sensitivity to environmental parameters, and nonlinearity.

6. Conclusions

In this paper, the long-term wind and wind effect BHMS data of an in-service prototype bridge are collected and sorted out. The essential characteristics of the VIV of the prototype bridge are discussed from a statistical perspective, and some important phenomena and conclusions different from wind tunnel experiments are discussed:

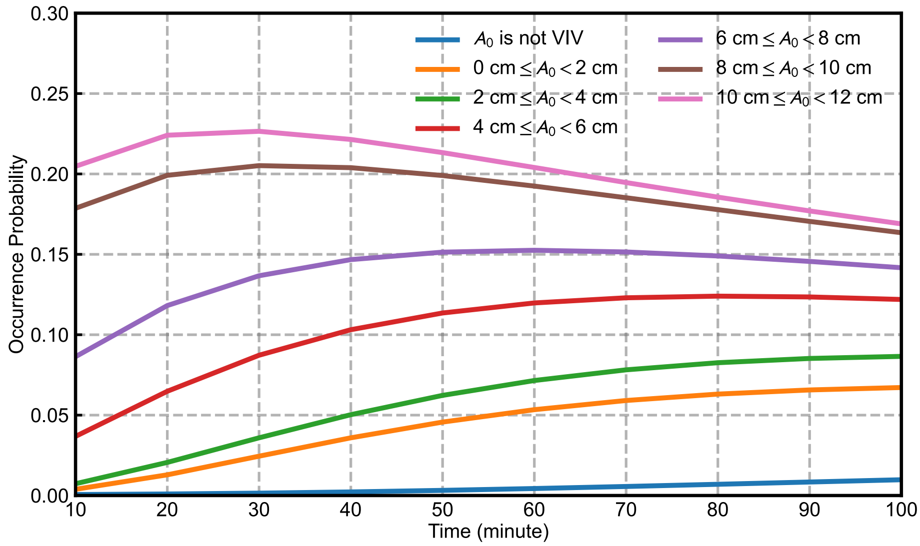


Fig. 24. Occurrence probability of time series.

- *Stochasticity*: Due to the influence of stochastic ambient excitations and the instability of natural wind field, the evolution of VIV amplitude of real bridges often does not conform to the classical theoretical formula. It seems unrealistic to establish a deterministic function relationship from observable parameters to the appearance or absence of VIV. Whether VIV occurs and the amplitude of VIV shows a clear probability distribution, and VIV samples and non-VIV samples often occur alternately.
- *Multi-mode*: The actual bridge may exhibit several VIV modes at the same wind speed. Even when there is not a noticeable change in wind speed during the process of VIV, the VIV mode may nevertheless shift from one to another.
- *Timescale-dependent*: Latent variables such as traffic flow and damping ratio, which are difficult to be directly observed, make VIV profoundly altered by both the daily and yearly time scales. For the investigated bridge, the traffic flow has an inhibitory effect on the VIV, and VIV is relatively less likely to occur in winter. Therefore, the assessment of VIV on the basis of short-term field measurements less than one year would be affected by systematic errors, and the long-term data can better reflect the number, mode and amplitude of the VIV in a comprehensive and accurate manner.
- *Markovianity*: By assuming the Markov property of the adjacent 10-min samples, a probability-based VIV prediction framework is proposed to assess the risk of amplitude exceeding a given threshold. The framework can take into account the vibration information of the previous time step and gradually evaluate the evolution of VIV probability at each time step.

CRediT authorship contribution statement

Liutian Zhang: Writing – original draft, Methodology, Formal analysis. **Wei Cui:** Writing – review & editing, Funding acquisition. **Lin Zhao:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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